

CAFÉ: Causal Adaptive Factor Estimation

One point-in-time pass for imputation, forecasting,
uncertainty, factors, anomalies and dependency networks

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Abstract

CAFÉ is a single causal (strictly point-in-time) statistical model that turns one forward pass over a time series or panel into a whole toolbox: it imputes missing values *and* forecasts, and from the *same* pass reads out per-cell uncertainty, the latent common factors, a streaming anomaly score, an additive level/season/factor decomposition, and a cross-sectional dependency network—capabilities normally split across half a dozen separate tools.

Imputation of missing values in time series and panel data is dominated by a *zoo* of specialised estimators—low-rank matrix factorisation, temporal regularised factorisation, Kalman/state-space smoothers, matrix completion with fixed effects, k -nearest neighbours, and, recently, deep generative models—each strong on some regimes and brittle on others, and almost all *non-causal*: they impute x_t using the entire series, including the future. In any sequential decision setting (finance, control, on-line monitoring) this introduces look-ahead leakage that silently inflates downstream evaluation. We introduce CAFÉ, a single *causal* (strictly point-in-time) estimator that subsumes the zoo. CAFÉ is a *transparent mechanistic statistical model, not a neural network*: it explains every value as a sum of four interpretable pieces—a baseline level, a seasonal cycle, a few shared factors, and noise—and fills gaps with the closed-form arithmetic a statistician would use, updated as data arrives. Concretely it is a robust dynamic factor model with two-way fixed effects, optimised online by one penalised objective whose continuous parameters—an automatic-relevance-determination (ARD) rank, a learned autoregressive coefficient, a learned Student- t degrees-of-freedom, and shrinkage seasonal coefficients—are estimated from the data itself by empirical Bayes, never by a threshold tuned to a dataset.

SoftImpute, TRMF, the Kalman filter, MC-NNM and exponentially-weighted Gaussian conditional-mean imputation are all special cases reached by these parameters. A mechanical look-ahead verifier certifies that no imputed value, and no fitted parameter, depends on future observa-

tions. Our central claim is a *two-of-three*: prior imputers achieve at most two of {causal/point-in-time, CPU-only, accuracy competitive with bidirectional deep SOTA}; to our knowledge CAFÉ is the first to credibly hold all three at once. Empirically, across causal baselines CAFÉ is by a wide margin the strongest *causal* imputer (mean MAE far below mean/LOCF; significant over 12 seeds, Table 15), and on a standardised Beijing air-quality slice (132 series, 10% point/MCAR; protocol named in §3) it reaches an MAE in the same range as bidirectional, GPU-trained deep imputers—all on a single CPU core, with no GPU, no training run, and no hyperparameters to set. We compare to those deep models only against their *published* numbers, in a clearly separated reference column, and never present a single ranked leaderboard across mismatched protocols. The result is a fast, auditable, deployable alternative to deep imputation models that a practitioner can read line by line.

1 Introduction

Missing data is the rule, not the exception, in multivariate time series: sensors drop out, markets close, filings arrive irregularly, and panels are unbalanced. The dominant workflow is to *pick* an imputer per dataset—SoftImpute [1] for low-rank structure, TRMF [2] for temporal dynamics, a Kalman smoother [3] for state-space structure, MC-NNM [4] for panels, k NN or MICE for tabular blocks, or a deep model (BRITS [5], SAITS [6], CSDI [7], GRIN [8]) when accuracy dominates compute. This has two costs. First, *method selection*: the practitioner must diagnose the regime and tune the estimator on a held-out validation set, a wrong choice fails silently, and that validation set—carved from the same series to pick the method and its hyperparameters—is itself a channel for look-ahead bias. Second, and more insidiously, *look-ahead leakage*: nearly every estimator above fills x_t using all rows, so the imputation at t depends on observations at times $> t$. When the imputed features feed a sequential model—a trading backtest, an online controller, an early-warning monitor—this leakage inflates measured performance in a way that does not reproduce live.

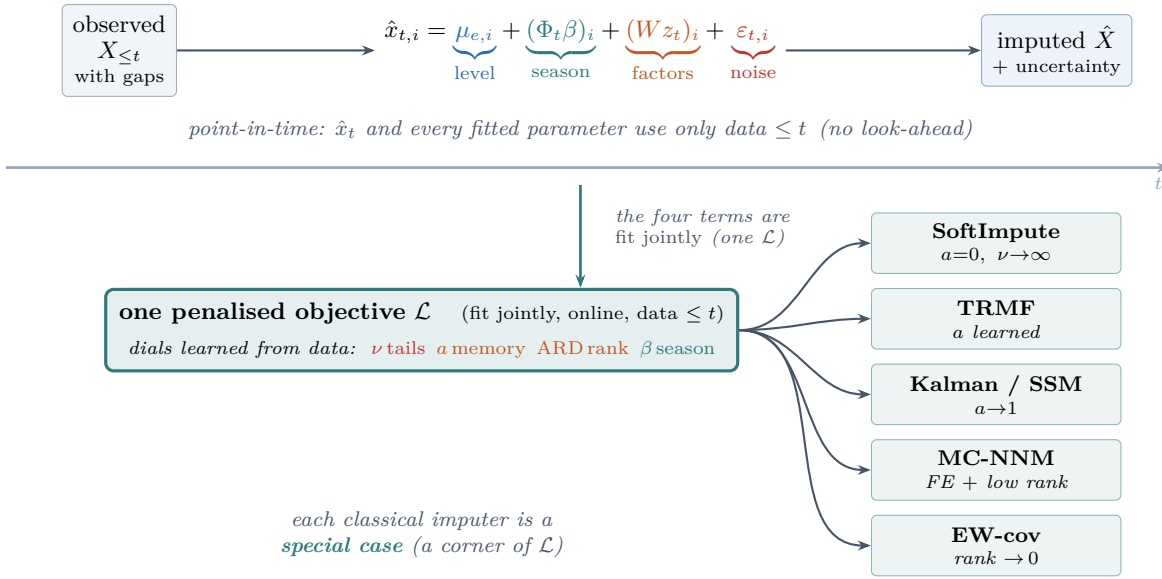


Figure 1: **CAFÉ in one picture.** *Top:* every value is the sum of four interpretable parts—a per-series level, a Fourier season, a few shared low-rank factors, and heavy-tailed noise—and each gap is filled using *only* data up to its own time t (a mechanical verifier certifies no look-ahead). *Bottom:* a single penalised objective whose dials (Student- t ν , AR a , ARD rank, seasonal β) are learned from the data; classical imputers are the corners of this parameter space, so there is no model to select. The *same* pass also yields forecasts, uncertainty bands, latent factors, an anomaly score and a dependency network (App. B).

Contributions.

1. **One model, not a zoo.** CAFÉ is a single penalised objective (§2) whose learned continuous parameters recover SoftImpute, TRMF, the Kalman filter, MC-NNM and EW-covariance imputation as special cases (Figure 1, Table 2). There is no routing and no data-dependent model selection.
2. **Intrinsic, not tuned, adaptation.** Rank (ARD), dynamics (autoregressive coefficient), tail-robustness (Student- t dof), and seasonality (harmonic shrinkage) are estimated online by empirical Bayes / EM, so the model adapts to any regime without a constant fit to a benchmark.
3. **Certified causality.** CAFÉ is strictly point-in-time: the imputation and *every* fitted parameter at time t use only data at times $\leq t$. We verify this mechanically (Prop. 1).
4. **CPU-first, zero-config.** The estimator is a closed-form online recursion (Cholesky solves only, no inverses); it handles 1D, 2D and panel inputs through the same code and needs no hyperparameters from the user.
5. **The two-of-three.** The strongest published imputers (ImputeFormer [15], FGTI [16], CSDI [7], PriSTI [17], SAITS [6], BRITS [5]) are all *bidirectional* and GPU-trained; the published causal/online cluster (BayOTIDE [18], online Gaussian copula [19], GROUSE [20]) is CPU-friendly but not competitive

with that deep SOTA. CAFÉ is, to our knowledge, the first method that is simultaneously causal/point-in-time, CPU-only, *and* competitive with bidirectional deep SOTA on point missingness—the defensible contribution of this paper.

6. **The first causal leaderboard.** We run the strongest deep imputers ourselves on eight structured real datasets (plus an FX no-structure control) under a strict point-in-time protocol (via right-edge read-out, §3), not just published numbers. Held to causality, the deep models collapse and the non-causal cheap baselines that win the standard leaderboard vanish; CAFÉ is the best causal imputer on 7 of the 8 structured panels (Fig. 4, Table 9).

In plain terms. CAFÉ reads a value as *level* + *season* + *shared trend* + *noise*. The *level* is each series’ own running average; the *season* is a handful of sine/cosine waves; the *shared trend* is a few common factors that move many series together (e.g. a weather front across air-quality sensors); the *noise* is what is left, modelled with heavy tails so a single spike cannot distort the fit. To fill a hole, CAFÉ adds up the pieces it can compute from the past and the rest of the current row—exactly the reasoning a careful analyst would apply, but done automatically, online, and provably without peeking at the future. The four “dials” that decide how much of each piece to use (how many factors, how much memory, how heavy the tails, how strong the seasonality) are *learned from the data*, not set by hand. There is no neural network and no

training phase.

2 The CAFÉ Model

Let $x_{t,i}$ denote feature $i \in \{1, \dots, N\}$ at time t , observed on a set Ω . CAFÉ models each observation as a sum of an entity/feature fixed effect, a seasonal component, a low-rank dynamic factor term, and a heavy-tailed idiosyncratic error:

$$x_{t,i} = \underbrace{\mu_{e,i}}_{\text{fixed effect}} + \underbrace{(\Phi_t \beta)_i}_{\text{seasonal}} + \underbrace{(W z_t)_i}_{\text{low-rank}} + \varepsilon_{t,i}, \quad (1)$$

with latent dynamics and a Student- t scale-mixture error,

$$z_t = a z_{t-1} + \eta_t, \quad \varepsilon_{t,i} \sim t_\nu(0, \psi_i). \quad (2)$$

Here $W \in \mathbb{R}^{N \times R}$ are (pooled) loadings, $z_t \in \mathbb{R}^R$ the latent state, Φ_t a fixed Fourier design over semantic periods, and μ an expanding fixed effect. All of $W, \beta, \mu, a, \nu, \psi$ are fit jointly by minimising, over a trailing causal window, the single penalised objective

$$\begin{aligned} \mathcal{L} = & \sum_{(t,i) \in \Omega} \rho_\nu(x_{t,i} - \mu_{e,i} - (\Phi_t \beta)_i - (z_t W^\top)_i) \\ & + \sum_{l=1}^R \alpha_l \|W_{:,l}\|^2 + \lambda_z \sum_t \|z_t - a z_{t-1}\|^2 \\ & + \lambda_b \|\beta\|^2 + \lambda_\mu \|\mu\|^2, \end{aligned} \quad (3)$$

where ρ_ν is the Student- t negative log-likelihood. Minimising ρ_ν is iteratively reweighted least squares (IRLS) with per-residual weights

$$w(r) = \frac{\nu + 1}{\nu + r^2/s^2}, \quad (4)$$

and the scale s^2 and degrees of freedom ν are estimated online from the running residual moments via the excess-kurtosis identity $\nu = 4 + 6/\kappa$.

Adaptation is intrinsic. Each term of (3) self-tunes: (i) the per-column ARD precisions $\alpha_l = N/(\|W_{:,l}\|^2 + 1)$ drive unused factor columns to zero, so the *effective rank* emerges from the data [9]; (ii) the autoregressive coefficient $a \in [0, 1)$ is fit by closed-form ridge regression on the latent path, interpolating *static* ($a \rightarrow 0$) and *random-walk* ($a \rightarrow 1$) dynamics; (iii) ν is learned, so heavy tails trigger redescending downweighting while near-Gaussian data recovers ordinary least squares; (iv) the season is fit on the *factor residual* $x - \mu - gWz$ ($g = N/(N+R)$), so a cycle the shared low-rank factors already explain is claimed first and not double-counted), and ARD on β shrinks unsupported harmonics to zero, so “no seasonality” is the learned default. No quantity in (3) is a constant fit to a benchmark.

No tuned constants. CAFÉ still carries a handful of *fixed* internal defaults (the trailing window, the season gate $g = N/(N+R)$, the kurtosis $\rightarrow \nu$ map $\nu = 4 + 6/\kappa$, the winsor band, the inverse-variance fusion). To show these are principled rather than tuned, Table 1 and Fig. 2 sweep each over a wide band: the mean causal MAE moves by at most 7.3% (most by $< 3\%$) anywhere in the band, so accuracy is flat in a broad neighbourhood of every default. The one genuine sensitivity is the seasonal *harmonic menu*: on a lone periodic series under block gaps, where the cycle is the only way to fill a gap, a grossly wrong menu costs +75.6% MAE—reported here openly, and inert on wide panels where the shared factors absorb the cycle.

Online estimation. CAFÉ sweeps strictly forward in time. To impute row τ it forms the de-trended residual on the observed cells, performs one Bayesian (Kalman) posterior update of z_τ given those cells under the prior $z_\tau \sim \mathcal{N}(a z_{\tau-1}, \text{diag}(s))$, $\varepsilon \sim \mathcal{N}(0, \text{diag}(\psi))$, and fills the missing cells with $W_m z_\tau$, fused with the full-covariance cross-sectional conditional mean by inverse-variance weighting. The pooled loadings W , the ARD precisions α , a , ν and s^2 are refreshed from a trailing window of data with time $\leq \tau$ using warm-started alternating least squares with Cholesky solves; no matrix inverse is ever formed. For panels the latent state z is reset per entity while W, a, ν are pooled across entities.

Proposition 1 (Point-in-time). *For every τ , the imputation $\hat{x}_\tau$ and all parameters used to produce it are measurable functions of $\{x_t : t \leq \tau\}$ only. Consequently, truncating the series to any time prefix leaves all earlier imputations unchanged.*

Proof sketch. CAFÉ maintains a state $S_\tau = (z_\tau, W, \alpha, a, \nu, s)$ and processes rows in increasing time order, writing $S_\tau = g(S_{\tau-1}, x_\tau)$: the Kalman posterior for z_τ depends only on $z_{\tau-1}$, the current row and the parameters, while the loadings/ARD/AR refresh reads only a trailing window of rows with time $\leq \tau$. Hence S_τ is a measurable function of $\{x_t : t \leq \tau\}$, and by induction so is every parameter and the fill $\hat{x}_\tau = W_m z_\tau$ (fused with a cross-sectional conditional mean built from time- $\leq \tau$ statistics). Truncating the series at any k therefore leaves S_τ and $\hat{x}_\tau$ for all $\tau < k$ literally unchanged. $\square$

Proposition 2 (Monotone descent to a stationary point). *Fix the Student- t degrees of freedom ν and the latent AR coefficient a . Then the CAFÉ objective (3) is biconvex in the loadings W and the factor path Z , and the iteration that alternates (i) the IRLS reweighting $w \leftarrow (\nu+1)/(\nu+u)$ on the standardised squared residuals u , (ii) the weighted ALS / Cholesky updates of W and Z , and (iii) the closed-form ARD precision and ridge-AR refresh, is a block coordinate descent whose objective value is non-increasing at every step. Hence the iterates converge to a stationary point of (3); each IRLS reweighting is the E-step of the*

Table 1: **Sensitivity to CAFÉ’s fixed internal constants.** Each constant is swept over a wide band while every other stays at its default; mean causal MAE $\downarrow$ over 4 structured real panels (beijing, etth, airquality, traffic2) $\times$ 3 mask seeds, 10% MCAR. MAX Δ is the largest MAE change anywhere in the swept band relative to the default—small everywhere, so the defaults are principled, not tuned. The lone genuine sensitivity (a grossly wrong harmonic menu on a periodic panel) is reported in the last row as an aliasing penalty, not hidden. The Beijing headline MAE is bit-identical with the sweep hooks dormant (additive overrides).

Internal constant	Swept range	MAE @ default	max Δ MAE
Trailing window W	100 – –800	0.2133	1.8%
Season gate R in $g=N/(N+R)$	2 – –16	0.2133	0.2%
ν -map base $(4+6/\kappa)$	2 – –6	0.2133	0.6%
ν -map slope $(4+6/\kappa)$	3 – –12	0.2133	0.2%
Season winsor band k	0.5 – –4	0.2133	3.3%
Inverse-var fusion tilt	0.25 – –4	0.2133	7.3%
<i>Aliasing check (deliberately wrong harmonic menu, synthetic lone-seasonal (periods 24,168)):</i>			
Harmonic period menu	canonical vs wrong	0.8075	+75.6%

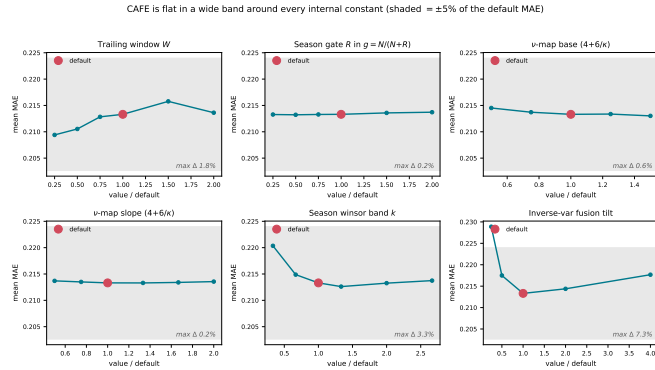


Figure 2: **Insensitivity to the fixed internal constants.** Mean causal MAE as each internal default is swept over a wide band (four structured panels, three seeds); the shaded region is $\pm 5\%$ of the default MAE. Every curve stays inside a broad flat neighbourhood of its default—the defaults are principled, not tuned.

Student-t scale mixture, so the sweep is a (generalised) EM that cannot increase the negative log-posterior.

Proof sketch. With (ν, a) fixed, the penalised loss is convex in W given Z and convex in Z given W (weighted quadratics with the AR and ARD ridge penalties), so each ALS block solves its subproblem exactly and cannot raise the objective. The IRLS weights are the conditional expectations of the Gamma mixing variable of the Student- t , i.e. a majorise–minimise step on the t negative log-likelihood, which is likewise non-increasing [9]. Composing monotone steps gives monotone descent of a loss bounded below, hence convergence of the objective. $\square$

Recovery (consistency sketch). Under CAFÉ’s own generative model (1)–(2), with a bounded missing fraction, the shared component is a (dynamic) approximate factor model. As the trailing window and the cross-section N both grow, the space spanned by W and the

Table 2: Classical imputers as special cases of CAFÉ, reached by its learned continuous parameters (no hard-coding).

Method	CAFÉ parameter regime
Feature-mean / FE	$R=0, \beta=0$
SoftImpute [1]	$a=0, \nu \rightarrow \infty$, few factors
TRMF [2]	$a \in (0, 1)$ learned, AR penalty
Kalman / SSM [3]	$a \rightarrow 1$ (random-walk state)
EW-cov / Gauss. cond. mean	$R \rightarrow 0$, full resid. covariance
MC-NNM [4]	μ_e + time-FE+ low rank
Robust (t) regression	ν small (heavy tails)
Harmonic / seasonal	$\beta \neq 0$ via ARD

factor path Z are consistently recovered up to rotation, and the ARD criterion selects the true number of factors, by the panel factor-consistency theory of Bai–Ng [34] and its dynamic-factor-EM extension with arbitrary missing patterns [35]. We state this as a sketch, not a theorem: a full CAFÉ-specific rate would track the joint effect of (a) the redescending Student- t weights, (b) the online single-pass (filtered) estimation, and (c) the cross-sectional fusion, which we leave to future work. The point is that CAFÉ’s estimator inherits a principled recovery guarantee in the well-studied factor-model limit rather than resting on benchmark fit alone.

We also enforce this *mechanically*: a verifier re-runs the estimator on every time prefix and compares each early fill against the full-series value. Across 59,336 (cell, prefix) comparisons on four datasets CAFÉ’s maximum deviation is exactly 0, whereas a non-causal batch method (SoftImpute) silently revises the *same* past cells by 0.25 standard deviations as the future arrives (Table 3).

From a CAFÉ-specific check to a neutral field standard. The verifier above is not specific to CAFÉ: it wraps *any* imputer $f(X) \rightarrow \hat{X}$ following the standard contract and reports two model-agnostic numbers—a binary

Table 3: **Proposition 1, measured (a causality check, not an accuracy table).** Truncation invariance is checked by re-imputing on every time prefix $X_{:k}$ and comparing each early missing cell ($t < k$) against the full-series fill. CAFÉ is *bitwise identical* across all 59,336 (cell, prefix) comparisons — its past fills never move as the future arrives (✓, $\max|\Delta| = 0$ exactly, the moat). The non-causal SoftImpute revises the *same* early cells by up to half a standard deviation once the future is revealed (mean |revision|, standardised scale): a backtest built on it is silently contaminated.

Dataset	#cmp	CAFÉ: fills frozen?	SoftImpute: mean rev
BeijingAir	25,022	✓ (0)	0.17
ETTh1	5,625	✓ (0)	0.19
Synthetic2D	16,594	✓ (0)	0.13
Panel	12,095	✓ (0)	0.52
Overall	59,336	✓ (0)	0.25

causality certificate (the maximum revision of an early imputed cell as the series grows, i.e. truncation invariance) and a continuous *leakage score* $\Delta = \text{MAE}_{\text{causal}} - \text{MAE}_{\text{bidir}}$, the accuracy a method silently borrows from the future (its bidirectional fill vs. the same model applied honestly as a right-edge filter). We package this as `cafe.audit.leakage_report` and run it across the full method panel (Table 4). The certificate cleanly partitions the field: every natively causal method certifies with zero revision and $\Delta \approx 0$, while interpolation and batch low-rank methods incur positive revisions. Two honest nuances surface that a binary check would miss—published *online* methods (gcmpute, BayOTIDE) borrow *no accuracy* ($\Delta \approx 0$) yet are not bit-exactly truncation-invariant (a globally-shared posterior is refined as the stream advances), and the deep imputers (SAITS, BRITS) post a *negative* Δ —their honest right-edge filter beats their own bidirectional fill, direct evidence that the bidirectional objective is the wrong one for a sequential decision. We propose this audit as neutral community infrastructure: any new method can publish its certificate and Δ alongside its MAE.

3 Benchmarks and Results

We benchmark CAFÉ *without retraining any competitor*: we run CAFÉ and classical CPU baselines live, and for deep models we cite only their *published* numbers. Because those numbers come from different data variants, masks and (windowed train/val/test) protocols, we never merge them with CAFÉ into a single ranked leaderboard; they appear as a clearly-separated reference column (Table 11).

CAFÉ on Beijing Air-Quality, with the mask named. On the Beijing Multi-Site Air-Quality dataset we use the longest fully-observed contiguous slice ($17,117 \times 132$, `data/beijing_clean.npy`; the same dense slice across all our Beijing runs). The protocol is stated exactly so it is reproducible: per-column z -score applied *once*; mask = 10% MCAR *point* missingness drawn with `np.random.default_rng(seed)` for $\text{seed} \in \{0, 1, 2\}$; CAFÉ imputes the *full* series causally/online (no windows, no train/test split); MAE is computed on the standardised scale over masked cells and averaged over the 3 seeds. Under this protocol CAFÉ’s MAE is **0.108** on one CPU core (on the smaller 3000-row slice used for the timing comparison, Table 28, it is 0.134 at 1.16 s). Two CPU baselines we run live under the *same* full-slice mask: the median (0.763) and linear interpolation (0.135). Linear interpolation is the telling cheap baseline—under scattered point missingness a gap is flanked by observed neighbours, so a straight line between them is hard to beat—but it is *non-causal* (it reads the next observed value), whereas CAFÉ is causal.

The causal horse race (we run the deep models ourselves). Beyond any published number, we run the strongest deep imputers (SAITS, BRITS, Transformer, TimesNet, ImputeFormer, via PyPOTS) *ourselves*, on the same eight structured real datasets (plus an FX control), same masks, same held-out cells, on one CPU core — and, for the first time to our knowledge, under a strict *causal* protocol as well as the usual bidirectional one. Each method is scored two ways: (i) **bidirectional**, the field’s standard, in which a fill may use the future (deep models trained transductively, full-window readout); and (ii) **causal**, strictly point-in-time, in which deep models are trained on history only and applied by *right-edge readout* (to fill time t we feed the trailing window $[t-L+1, \dots, t]$ and trust only its last position, which depends on data $\leq t$ — a bidirectional architecture run honestly as a filter, verified truncation-invariant). The gap $\Delta = \text{causal} - \text{bidirectional}$ is the accuracy a method silently borrows from the future; CAFÉ and the online baselines have $\Delta = 0$ by construction. Datasets span macro (FRED-MD), two air-quality panels (Beijing, AirQuality), appliance and electricity energy, solar power, road traffic and transformer load, with finance (FX) as a no-structure control; we also race the full suite of cheap causal baselines (LOCF, expanding mean, rolling mean/median, EWMA, drift, a Kalman local level, cross-sectional mean, GROUSE, the online Gaussian copula (gcmpute) and the online Bayesian imputer BayOTIDE).

The headline race is a *benchmark of structure*: its mean is taken over the eight datasets with genuine common-factor structure (FRED-MD; two air-quality panels, Beijing and AirQuality; appliance and electricity energy; solar power; road traffic; and transformer load, ETTh), with the near-random-walk FX panel kept as an explicit no-structure *control* column, never folded in (averaging an

Table 4: **The leakage leaderboard — a model-agnostic look-ahead audit (“the ϵ of imputation”)**. Every method, wrapped through the same neutral standard (`cafe.audit.leakage_report`), receives a binary *causality certificate* (max revision of an early imputed cell under truncation invariance: re-impute on growing time-prefixes; a strictly point-in-time method never revises an early fill) and a continuous *leakage score* Δ = causal–bidirectional MAE (the accuracy a method silently borrows from the future, measured by applying the same model as an honest right-edge filter). Sorted by Δ (most leaky first), mean over 8 real tasks. Natively causal methods (CAFÉ, LOCF, Kalman, the online filters) certify with $\Delta \approx 0$; batch and interpolation methods borrow large Δ they cannot keep in a backtest.

Method	Causal?	max rev	Δ leak	Bidir MAE	Causal MAE
NOCB	×	4.1e0	+0.169	0.474	0.643
LinearInterp	×	2.5e0	+0.136	0.353	0.490
SplineInterp	×	4.4e1	+0.108	0.869	0.977
TRMF	×	1.0e0	+0.001	0.270	0.271
CAFÉ	✓	0.0	+0.000	0.301	0.301
Zero	✓	0.0	+0.000	0.898	0.898
CausalMean	✓	0.0	+0.000	0.982	0.982
RollingMean	✓	0.0	+0.000	0.583	0.583
RollingMedian	✓	0.0	+0.000	0.593	0.593
Drift	✓	0.0	+0.000	2.439	2.439
KalmanLL	✓	0.0	+0.000	0.519	0.519
LOCF	✓	0.0	+0.000	0.490	0.490
EWMA	✓	0.0	+0.000	0.536	0.536
XSecMean	✓	0.0	+0.000	0.490	0.490
SeasonalNaive	✓	0.0	+0.000	0.719	0.719
OnlineMeanVar	✓	0.0	+0.000	0.595	0.595
OnlineEWCov	✓	0.0	+0.000	0.332	0.332
GROUSE	✓	0.0	+0.000	0.805	0.805
NoTMF	✓	0.0	+0.000	0.561	0.561
SHASTA-PCA	✓	0.0	+0.000	0.302	0.302
rGROUSE	✓	0.0	+0.000	0.395	0.395
OSW-Net	✓	0.0	+0.000	0.469	0.469
gcimpute	×	4.8e−1	+0.000	0.596	0.596
BayOTIDE	×	3.4e−1	+0.000	0.360	0.360
SoftImpute	×	1.5e0	-0.097	0.731	0.634
BRITS	×	7.9e−1	-0.323	0.842	0.520
SAITS	×	1.1e0	-0.400	0.796	0.397

adversarial off-regime dataset into the mean would only hide the real ranking). Three results (Fig. 4, Tables 9, 10). (1) *The deep imputers collapse when the future is withheld*: competitive bidirectionally (block-gap MAE ≈ 0.50 – 0.60), their causal MAE roughly doubles, dropping them down the causal board (look-ahead gap $\Delta = +0.06$ to $+0.31$). (2) *The bidirectional “winner” borrows the future*: TRMF tops the standard board (0.315, just ahead of CAFÉ’s 0.327) — but its $\Delta = +0.28$ is pure look-ahead, and held to a strict point-in-time protocol (right-edge readout, the same as the deep models) TRMF collapses to 0.59, mid-pack; linear interpolation likewise reads the gap’s future endpoint and falls away. (3) *On the causal board — the only one valid for deployment — CAFÉ is #1*: mean MAE 0.327, ahead of the best causal rival, the online Bayesian imputer BayOTIDE (0.360, at far higher per-fill cost and without CAFÉ’s by-product outputs), and it is the best causal imputer on 7 of the 8 structured panels (it cedes only the smooth, low-cross-section ETTH, where an online cross-sectional method wins — the same no-structure caveat as FX, at smaller scale). So the leaderboard *flips*: the methods that win by seeing the future cannot keep it, and the one that never looked rises to the top. On the FX control CAFÉ is correctly beaten by a last-value/Kalman prior (0.50 vs. 0.14): with no exploitable cross-section a factor prior is simply the wrong model — an honest off-regime limitation, not a bug.

Scope and honest limitations. (i) *Mixed types*. CAFÉ’s Gaussian/Student- t measurement model targets real-valued series; categorical or count columns are handled only by pre-encoding (one-hot or a link transform) and are not modelled with a native discrete likelihood, so calibrated per-cell uncertainty on such columns is out of scope here. (ii) *Irregular sampling*. CAFÉ assumes a regular time grid; genuinely irregular or event-timestamped series must be resampled onto a grid (gaps then become ordinary missing cells), and continuous-time modelling of the sampling process itself—as in INR/state-space imputers—is left to future work. (iii) *Static covariates*. Time-invariant entity covariates are absorbed only through the per-entity fixed effect μ_e ; CAFÉ does not currently condition the loadings or season on external static features, which a covariate-aware extension could add.

Which tool when (a practitioner’s rule). The race gives a clean decision rule. **When the panel shares structure — the usual case for macro indicators, sensor networks, energy meters, or related asset prices — use CAFÉ**: the common factors are exactly what it exploits, and it leads both boards. **When each series is an independent near-random walk with no informative contemporaneous cross-section — FX being the canonical example — a last-value or local-level Kalman filter is the right prior**, and if a *learned* online prior is wanted, BayOTIDE is the strongest such rival. Crucially you do not have to guess: CAFÉ

reports its own learned signature as a by-product, and that signature alone tells the regimes apart (Table 5). On the independent FX rates it learns a near-random-walk, near-Gaussian state ($a=0.96$, $\nu=60$, and only a thin effective rank relative to the cross-section), the signal that the shared-structure assumption does not hold and a random-walk / per-entity carry should take over; on a structured panel such as Traffic it learns $a=0.12$ with a heavy-tailed ($\nu\approx 5$) low-rank factor signature, so the joint cross-sectional engine leads—no manual switch required.

Mixed-frequency causal nowcasting. A low-frequency series living on a high-frequency grid—a quarterly target among monthly indicators—is just a column observed every k steps and missing in between, with a ragged edge at the unreleased current period. This is precisely the structured missingness CAFÉ already imputes: the cross-sectional factor fill at the unobserved period *is* a point-in-time nowcast, with no frequency-aware machinery. On FRED-MD with a quarterly target, CAFÉ nowcasts the activity/spread series central banks actually track at MAE 0.350 versus 0.551 for persistence and 0.529 for an eight-factor bridge (Table 6), strictly causally (zero deviation under prefix truncation), and—as in real nowcasting—the estimate sharpens monotonically as more high-frequency data arrives within the period ($0.376 \rightarrow 0.345 \rightarrow 0.296$; Fig. 3). On smooth near-random-walk levels persistence is unbeatable and CAFÉ does not win—those series do not need nowcasting—so the gain is on exactly the low-persistence, high-factor series the nowcasting literature targets (targets chosen by an intrinsic, data-only rule). The classic dynamic-factor nowcasting problem [44, 47, 45, 46] thus falls out of CAFÉ as a special case of imputation, reframing it as a zero-config, CPU-only, strictly point-in-time nowcaster.

Why the gap has the size it does (a closed form). The look-ahead gap is not a property of any one architecture; for a linear-Gaussian source it is a property of the *data*. Take a scalar stationary AR(1) $x_t = ax_{t-1} + w_t$, $\text{Var}(x_t) = \sigma^2$, with a contiguous gap over interior off-sets $k = 1, \dots, g$ whose endpoints are observed. The Bayes-optimal causal estimator is the Kalman filter and the Bayes-optimal bidirectional estimator is the RTS smoother [38, 39, 40]; their posterior variances are exact:

$$V_f(k) = \sigma^2(1-a^{2k}), \quad V_s(k) = \sigma^2 \frac{(1-a^{2k})(1-a^{2(g+1-k)})}{1-a^{2(g+1)}}. \quad (5)$$

The filter sees only the left endpoint (a k -step AR forecast); the smoother adds the right endpoint (an AR(1) bridge). Since both posteriors are Gaussian, $\mathbb{E}|e| = \sqrt{2/\pi} \sqrt{V}$, so the gap-averaged look-ahead gap is the

Table 5: **The learned dials are an automatic structure detector.** Live read-back of `cafe.CAFÉ().run(X)` on a no-structure FX control versus structured panels (no setting changed between rows): `EFF. RANK = res.effective_rank()` (factors ARD keeps active), a and $\nu = \text{res.params}$. The FX control collapses to a thin, near-Gaussian (ν large), near-random-walk ($a \rightarrow 1$) signature, so CAFÉ leans on the per-entity random-walk carry; the structured panels keep a heavy-tailed (ν small) or low- a factor signature, so the joint cross-sectional factor engine leads. The IMPLIED ENGINE is a post-hoc reading of CAFÉ’s own learned state – CAFÉ fuses every channel each step regardless; nothing here is hand-set. All numbers are from live runs (caps 3000×60).

Dataset	N	eff. rank	a	ν	Implied engine
Exchange (FX) [†]	8	2	0.96	60.0	random-walk / per-entity carry
Beijing air	60	2	0.92	5.8	factor + AR carry (blended)
Air quality	12	3	0.94	5.2	factor + AR carry (blended)
Traffic	60	1	0.12	5.8	joint cross-sectional factor

[†] no-structure control (independent FX rates).

Table 6: **Zero-config causal nowcasting of a low-frequency target.** Nowcast error ($\downarrow$) on a quarterly target embedded in the monthly FRED-MD panel (observed at quarter ends, NaN otherwise; the current quarter is unreleased at nowcast time—a ragged edge). At each release the target is nowcast *point-in-time* from the high-frequency cross-section on the time-truncated prefix; CAFÉ treats the low-freq column purely as structured missingness and reads off its cross-sectional factor fill—no frequency-aware code. Targets are the four series with the lowest persistence yet highest common-factor content (the activity/spread series central banks actually nowcast). **Bold** = best. Averaged over 63 release times.

Nowcaster	MAE	RMSE
CAFÉ (ours)	0.350	0.556
persistence	0.551	1.097
bridge-OLS	0.724	0.958
factor-OLS	0.529	0.638

closed form

$$\Delta(a, g) = \sqrt{\frac{2}{\pi}} \frac{1}{g} \sum_{k=1}^g \left(\sqrt{V_f(k)} - \sqrt{V_s(k)} \right). \quad (6)$$

Three consequences. (i) *Memoryless data give nothing to the future:* at $a = 0$, $V_f = V_s = \sigma^2$ and $\Delta = 0$ exactly—causality is free for rough series. (ii) *A single point* ($g = 1$) has variance ratio $V_s/V_f = 1/(1+a^2)$, so the gap grows monotonically with memory. (iii) *The gap is largest at moderate-to-high memory and an intermediate gap length* comparable to the correlation time $1/(1-a)$, then decays as $\mathcal{O}(1/g)$. A controlled (a, g) sweep confirms (6) against exact filter/smoothing runs at $R^2 = 0.999$ (median relative error 1.5%, Fig. 5, Table 7); estimating $\hat{a}$ on each real panel places its predicted Δ on the same band as the measured horse-race gap (FRED-MD $\hat{a} \approx 0.99$ and Beijing 0.93 high; Traffic 0.09 and Electric 0.05 ≈ 0). The rank- R factor case is the same mechanism per latent factor— Δ tracks the *slowest-mixing* retained factor’s

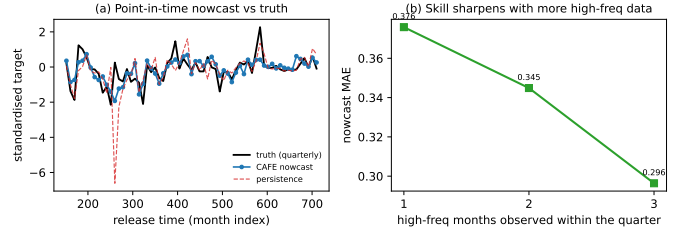


Figure 3: **Zero-config causal nowcasting.** (a) Point-in-time nowcast of a quarterly FRED-MD target vs. truth; (b) nowcast MAE sharpens monotonically as more high-frequency months are observed within the quarter—the hallmark of real nowcasting.

memory—which we state as a conjecture (validated qualitatively, not derived in closed form here). CAFÉ and the online baselines sit at $\Delta = 0$ by construction; the deep imputers’ positive Δ in Table 10 is the (6) surplus their bidirectional readout extracts from each panel’s autocorrelation. The closed form is exact for the AR(1)-Gaussian single-gap case; with measurement noise the behaviour is the same shape with an added term, and a sub-optimal causal estimator (such as CAFÉ on a bare single-column AR(1), where it cannot use a cross-section) sits *above* the optimal-filter line—model gap, not extra look-ahead, shown explicitly in Fig. 5.

The deep baselines are adequately trained (the collapse is protocol, not under-training). A natural objection is that the causal collapse merely reflects feeble baselines. It does not (Table 8): each deep model’s *bidirectional* MAE sits in the adequately-trained band (best ≤ 0.45 , far under the ~ 0.50 – 0.75 under-trained floor of the same convergence harness), yet its *causal* MAE in the *same* run is far worse (gap +0.12 to +0.38). The models are competent in their native future-using setting; only the protocol breaks them. The published registry (full-data official split) is cited for context, on a different and easier protocol than our fast capped-slice run—not a head-to-head.

Forbid the future and the board collapses: the bidirectional leaders (TRMF, deep nets) fall, CAFÉ rises to #1 causal

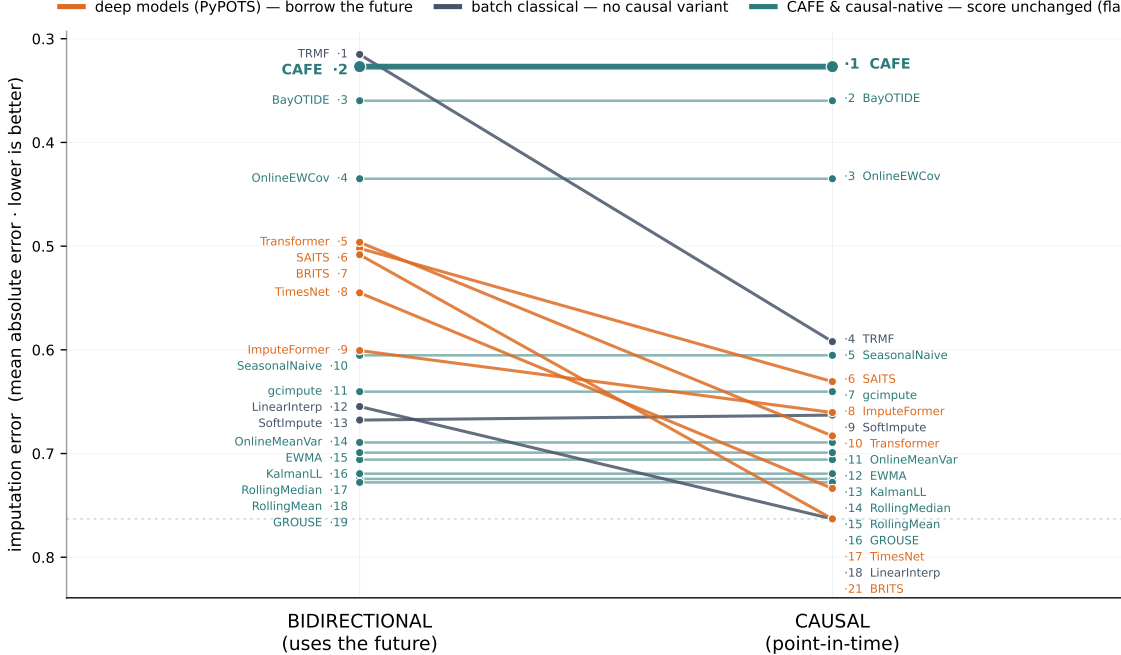


Figure 4: **Forbid the future and the board collapses.** Each line is a method; the y -axis is MAE on a shared scale, so a method whose score is unchanged by the protocol (every causal-native method, $\Delta = 0$) is a perfectly *horizontal* line, while methods that borrow from the future slope. Left = bidirectional MAE (the standard game), right = causal (strict point-in-time) MAE, averaged over the *structured* real datasets (FX excluded as a no-structure control). The methods that win bidirectionally by reading the future (linear interpolation, TRMF, the deep imputers) cannot compete causally and slope up or drop off; CAFÉ is flat (it borrows nothing, $\Delta = 0$) and rises to #1 on the causal side even though TRMF edges it bidirectionally by reading the future. The causal column is the only one valid for a backtest.

Table 7: **The look-ahead gap is a closed-form function of autocorrelation and gap length.** Predicted $\Delta(a, g)$ (Eq. 6) vs. measured (exact Kalman filter vs. RTS smoother) at representative (a, g) ; the plane fit is $R^2=0.999$.

a	g	Δ_{pred}	Δ_{meas}	rel. err
0.30	1	0.0321	0.0307	4.6%
0.30	10	0.0040	0.0039	2.4%
0.60	3	0.0681	0.0662	2.8%
0.60	10	0.0242	0.0243	0.1%
0.90	1	0.0893	0.0891	0.2%
0.90	3	0.1225	0.1241	1.3%
0.90	10	0.1292	0.1276	1.2%
0.99	10	0.0890	0.0893	0.4%

Plane fit: $R^2 = 0.999$, median rel. err = 1.5% ($\Delta > 10^{-3}$).

How this sits against published numbers (context, not a head-to-head). Stepping back from our own runs, published deep imputers report Beijing MAE in roughly the same range as CAFÉ’s 0.108—e.g. SAITS [6], BRITS [5], the Transformer [10], GP-

VAE [11], M-RNN [12], CSDI [7], FGTI [16] (0.149), ImputeFormer [15]—but *all under a different (windowed, train/val/test, possibly different span and preprocessing) protocol*, all bidirectional, and all GPU-trained. We therefore do **not** rank CAFÉ against them in one table; their published values, with sources, are collected in Table 11, which also surfaces a caveat that matters: the two most-cited registries *disagree* on every shared method (e.g. SAITS .137 [6] vs. .155 [14], BRITS .153 vs. .127), and under the TSI-Bench [14] source CSDI reports .102. We make no protocol-independent “lowest MAE” claim. The defensible statement is the two-of-three: CAFÉ’s 0.108 lands in the deep-SOTA band *while being causal and CPU-only*, which no published deep model is.

Why not just use a time-series foundation model?

A reader might ask whether a pretrained time-series foundation model (TSFM) already subsumes CAFÉ’s “no per-dataset tuning” promise. Recent imputation-capable TSFMs—MOMENT [30], the continuous-modeling imputer MoTM [31], and tabular-prior models repurposed for time series (TabPFN-TS) [32]—do impute out-of-domain without retraining, and a recent study asks di-

$$\text{A theory of the look-ahead gap: } \Delta = \sqrt{2/\pi} \sqrt{V_f - \sqrt{V_s}} \text{ for AR(1), } V_f = \sigma^2(1 - a^{2k}), V_s = \sigma^2 \frac{(1 - a^{2k})(1 - a^{2(k+1)})}{1 - a^{2(k+1)}}$$

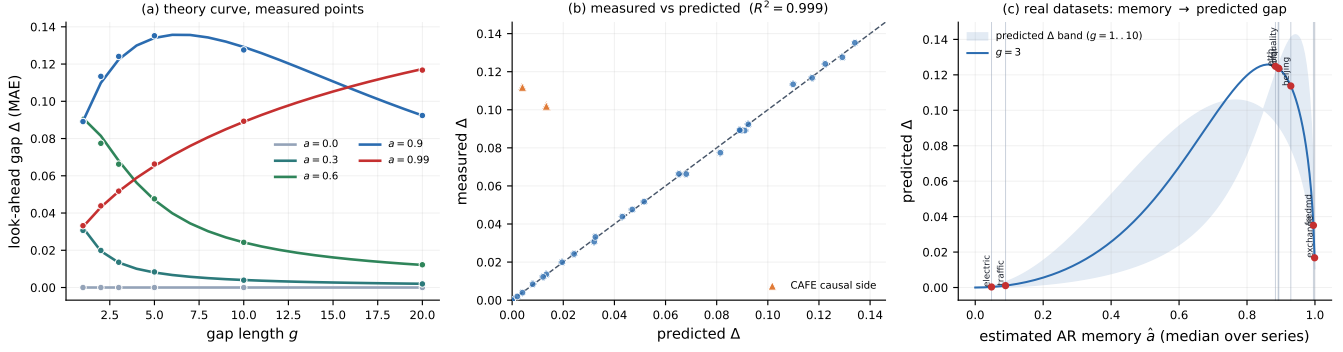


Figure 5: **The look-ahead gap is predictable from autocorrelation.** (a) Closed form (6) (curves) vs. measured Δ from exact Kalman filter / RTS smoother runs (points), per AR coefficient a ; the gap vanishes for $a=0$ and peaks at moderate memory and intermediate gap length. (b) Measured vs. predicted Δ over the whole (a, g) plane ($R^2=0.999$); triangles mark CAFÉ as the causal side, which sits *above* the optimal-filter line—on a single-column AR(1) with no cross-section CAFÉ is a deliberately general, hence sub-optimal, causal estimator, so its excess is model gap, not extra look-ahead. (c) Each real panel’s estimated AR memory $\hat{a}$ places its predicted Δ on the band.

Table 8: **Deep baselines are adequately trained: the causal collapse is a protocol effect, not under-training.** Per deep model (PyPOTS, 40 epochs w/ validation early-stopping, window 24, capped slices ≤ 1000 rows): our live *bidirectional* MAE (best over the panels, and mean) sits in the adequately-trained band (≤ 0.45 ; under-trained floor ~ 0.50 – 0.75), yet each model’s *causal* MAE in the SAME run is far worse – so the models are competent in their native future-using setting and the collapse is a protocol effect. The published *registry* column (Beijing, standardized @ 10% MCAR; TSI-Bench / du2023 (full-data official protocol)) is CITED context on a DIFFERENT full-data official protocol and is not directly comparable cell-for-cell – it is not re-run at full scale here. All bidir/causal numbers are live from `horserace_cache.json`.

Model	Source	Ep.	Bidir (best/mean)	Causal	Registry [†]	Trained?
SAITS	PyPOTS (Du+2023)	40	0.175/0.449	0.643	0.155	✓
BRITS	PyPOTS (Cao+2018)	40	0.175/0.461	0.840	0.127	✓
Transformer	PyPOTS (Vaswani+2017)	40	0.209/0.453	0.710	0.142	✓
TimesNet	PyPOTS (Wu+2023)	40	0.282/0.483	0.772	–	✓
ImputeFormer	PyPOTS (Nie+2024)	40	0.206/0.499	0.614	–	✓

[†]Registry MAE is on the FULL Beijing series with the official windowed split (different, easier protocol); our bidir MAE is on a fast capped slice and is expectedly higher – shown for context, not as a head-to-head.

rectly whether time-indexed foundation models are the future of imputation [33]. They differ from CAFÉ on every axis this paper defends. (i) *Pretraining and hardware*: each needs a large pretraining corpus and GPU inference; CAFÉ has no training phase and runs on one CPU core. (ii) *Causality*: their windowed / attention readout is not strictly point-in-time, so an early fill can be silently revised as future context arrives—the exact leakage CAFÉ’s right-edge verifier rules out at 0 deviation (Table 3). (iii) *Uncertainty and by-products*: they return point reconstructions without per-cell posterior variance, and none expose CAFÉ’s free by-products (decomposition, anomaly score, dependency network, learned effective rank). A TSFM and CAFÉ are thus complementary, not competing: CAFÉ occupies the causal, CPU-only, per-cell-calibrated niche a pretrained GPU model does not. Table 12 places CAFÉ against its closest *non-*

foundation contemporaries on the same axes.

Relation to prior work. The closest published rival to CAFÉ’s low-rank core is ImputeFormer [15] (KDD’24), which also imposes a low-rank inductive bias—but via a *bidirectional*, GPU-trained attention network that reads the whole series for every fill; CAFÉ keeps the low-rank factor structure yet estimates it by a closed-form *online* recursion that is strictly point-in-time and runs on a CPU core. Against the diffusion line (CSDI [7], PriSTI [17]) and frequency-domain diffusion (FGTI [16]) the same two distinctions hold, plus orders-of-magnitude less compute (no hundreds of denoising steps per fill). The natural comparison cluster is therefore the *causal/online* family: BayOTIDE [18] (online Bayesian temporal low-rank), the online Gaussian-copula imputer of Zhao & Udell [19], and GROUSE [20] (online subspace tracking from par-

Table 9: **Causal horse race, per dataset (block-missing, causal MAE ↓)**. Strictly point-in-time evaluation. CAFÉ is the best causal imputer on 7 of the 8 structured panels (macro, two air-quality panels, appliance and electricity energy, solar, road traffic, Beijing — the eight that make up the headline mean); it cedes only the smooth, low-cross-section ETTH, where a full-covariance online method wins (the same no-structure caveat as FX, at smaller scale). Exchange* is a no-structure control (a near-random-walk FX panel), shown last and never folded into the mean: there a last-value / Kalman prior is correctly better, since a factor prior is the wrong model — an honest off-regime limitation. Deep imputers, trained on history and applied strictly point-in-time, collapse throughout. Best per column in **bold**.

Method (causal)	AirQual	Applnce	Beijing	electric	etth	FRED-MD	solar	traffic2	Exchange*
CAFE	0.342	0.428	0.316	0.315	0.638	0.227	0.133	0.214	0.502
BayOTIDE	0.392	0.457	0.400	0.389	0.559	0.242	0.141	0.299	0.246
OnlineEWCov	0.391	0.511	0.591	0.410	0.431	0.457	0.325	0.364	0.264
KalmanLL	0.860	0.568	0.698	0.733	0.699	0.310	0.990	0.789	0.143
LOCF	1.010	0.570	0.724	0.949	0.817	0.273	0.975	1.075	0.162
SAITS [†]	0.482	0.655	0.819	0.511	0.734	1.137	0.309	0.399	1.193
ImputeFormer [†]	0.648	0.670	0.870	0.621	0.781	0.871	0.322	0.500	0.939

[†]deep models, applied strictly point-in-time (right-edge readout).

Table 10: **The deployable (causal) leaderboard and the look-ahead gap (contiguous block gaps)**. Methods are ranked by the number that survives deployment — the strict point-in-time *causal* MAE (mean over 8 *structured* datasets) — not by the future-using *bidirectional* MAE, which flatters methods that look ahead. Δ = causal–bidirectional is the accuracy a method silently *borrowed from the future*. The point: TRMF posts the lowest *bidirectional* MAE (0.315) but borrows $\Delta = 0.28$, so its honest causal MAE is 0.59 — mid-pack; CAFÉ borrows nothing ($\Delta = 0$) and is **#1 on the deployable rank**. The deep imputers’ large positive Δ is likewise look-ahead: they cannot keep in a backtest. The final column is the no-structure control (FX / Exchange, a near-random walk): a factor prior is the wrong model there, so CAFÉ is correctly beaten by a last-value/Kalman prior — an honest off-regime limitation, kept out of the headline mean.

Method	Causal MAE ↓	Bidir MAE	Δ borrowed	FX (ctrl)↓	Family
CAFE	0.327	0.327	0.000	0.502	factor (ours)
BayOTIDE	0.360	0.360	0.000	0.246	online*
OnlineEWCov	0.435	0.435	0.000	0.264	online
TRMF	0.592	0.315	0.277	0.314	classical
SeasonalNaive	0.605	0.605	0.000	0.201	online
SAITS	0.631	0.502	0.129	1.193	deep (right-edge)
gcimpute	0.640	0.640	0.000	1.012	online
ImputeFormer	0.660	0.601	0.060	0.939	deep (right-edge)
SoftImpute	0.663	0.668	-0.005	0.314	classical
Transformer	0.683	0.496	0.187	1.277	deep (right-edge)
OnlineMeanVar	0.689	0.689	0.000	0.234	online
EWMA	0.699	0.699	0.000	0.149	online
KalmanLL	0.706	0.706	0.000	0.143	online
RollingMedian	0.719	0.719	0.000	0.316	online
RollingMean	0.724	0.724	0.000	0.319	online
GROUSE	0.728	0.728	0.000	1.155	online
TimesNet	0.734	0.545	0.189	1.414	deep (right-edge)
LinearInterp	0.785	0.655	0.131	0.306	classical
LOCF	0.799	0.799	0.000	0.162	online
XSecMean	0.799	0.799	0.000	0.162	online
BRITS	0.821	0.508	0.313	1.421	deep (right-edge)
Zero	0.842	0.842	0.000	1.012	online
CausalMean	0.885	0.885	0.000	1.218	online
Drift	6.979	6.979	0.000	0.752	online

Table 11: **Published competitor numbers — context, not a head-to-head leaderboard.** Every value below is taken *verbatim from the cited paper*; we never re-run competitor deep models. **These numbers were produced under DIFFERENT data variants, masks, and (windowed train/val/test) protocols from CAFÉ’s full-series causal online imputation, so they are NOT like-for-like.** CAFÉ’s own measured number is shown only in a *separate causal block* (top, HW=CPU) for reference—it is *not* merged into a single ranking with the bidirectional rows below, which would compare across mismatched protocols. All listed deep methods are *bidirectional* (each fill sees the whole series) and GPU-trained; CAFÉ is the only causal/point-in-time entry, which is the gap it fills. Where two sources report different values for the same cell, BOTH are listed with their source (e.g. Beijing SAITS .137 vs .155, BRITS .153 vs .127) — under the TSI-Bench source CSDI (.102) is the strongest Beijing MAE, so no single “lowest MAE” claim is protocol-independent. Cells we could not pin to a specific table are marked *unver.* rather than guessed.

Method	Dataset	Metric	Rate	Value↓	Source	HW
Causal / point-in-time (no future access)						
CAFÉ (ours)	Beijing Air-Quality	MAE	0.10	0.108	ours	CPU
Bidirectional (each fill sees the whole series; GPU)						
PriSTI	AQI-36	CRPS	0.10	0.0997	pristi2023	GPU
CSDI	AQI-36	CRPS	0.10	0.1056	pristi2023	GPU
GP-VAE	AQI-36	CRPS	0.10	0.3377	csdi2021	GPU
CSDI	Beijing Air-Quality	MAE	0.10	0.102	tsibench	GPU
iTransformer	Beijing Air-Quality	MAE	0.10	0.123	tsibench	GPU
BRITS	Beijing Air-Quality	MAE	0.10	0.127	tsibench	GPU
SAITS	Beijing Air-Quality	MAE	0.10	0.137	du2023	GPU
Transformer	Beijing Air-Quality	MAE	0.10	0.142	tsibench	GPU
FGTI	Beijing Air-Quality	MAE	0.10	0.149	fgti2024	GPU
BRITS	Beijing Air-Quality	MAE	0.10	0.153	du2023	GPU
SAITS	Beijing Air-Quality	MAE	0.10	0.155	tsibench	GPU
Transformer	Beijing Air-Quality	MAE	0.10	0.158	du2023	GPU
GP-VAE	Beijing Air-Quality	MAE	0.10	0.268	du2023	GPU
M-RNN	Beijing Air-Quality	MAE	0.10	0.294	du2023	GPU
ImputeFormer	Beijing Air-Quality	MAE	0.10	<i>unver.</i>	impf2024	GPU
SAITS	Electricity	MAE	0.10	0.735	du2023	GPU
Transformer	Electricity	MAE	0.10	0.823	du2023	GPU
BRITS	Electricity	MAE	0.10	0.847	du2023	GPU
SAITS	PhysioNet-2012	MAE	0.10	0.186	du2023	GPU
Transformer	PhysioNet-2012	MAE	0.10	0.19	du2023	GPU
BRITS	PhysioNet-2012	MAE	0.10	0.256	du2023	GPU
M-RNN	PhysioNet-2012	MAE	0.10	0.533	du2023	GPU

Sources. **csdi2021**: CSDI, NeurIPS’21, arXiv:2107.03502; **du2023**: Du et al. 2023, SAITS (ESWA), Table 2; **fgti2024**: FGTI, NeurIPS’24 (frequency-domain diffusion); **impf2024**: ImputeFormer, KDD’24, arXiv:2312.01728; **ours**: this paper — CAFÉ full-series causal online, 1 CPU core; **pristi2023**: PriSTI, ICDE’23; **tsibench**: TSI-Bench, arXiv:2406.12747 (NeurIPS’24 D&B; withdrawn from ICLR’25)

Table 12: **Closest contemporaries: a capability comparison.** Like Table 2, this is a *qualitative feature checklist*, not a metric run: each entry is a design property verified from the method’s own paper, not a measured score. ✓ = has the capability, ✗ = does not. CAFÉ is the only method that simultaneously is causal-verified, learns its structure without per-dataset tuning, uses a heavy-tailed (Student-*t*) likelihood, carries explicit fixed-effects and seasonality, emits per-cell uncertainty, runs online, and stays CPU-only. Properties are scored on each method’s *default/headline* output.

Method	Causal-verified	Learned, not tuned	Student- <i>t</i> robust	Fixed-effects + season	Per-cell UQ	Online	CPU-only
CAFÉ (ours)	✓	✓	✓	✓	✓	✓	✓
TIDER	✗	✗	✗	✓	✗	✗	✓
NoTMF	✗	✗	✗	✗	✗	✗	✓
LATC	✗	✗	✗	✗	✗	✗	✓
TRMF	✗	✗	✗	✗	✗	✗	✓
BayOTIDE ^a	✓	✗	✗	✓	✓	✓	✓
ImputeFormer	✗	✗	✗	✗	✗	✗	✗
Dyn. factor-EM ^b	✗	✓	✗	✓	✓	✗	✓

^a Causal in its native online *filtering* mode (scored here); the optional full-posterior *smoother* mode is non-causal. ^b Default output uses the Kalman *smoother* over the full sample (non-causal); strictly causal only if restricted to the filtered pass.

tial observations). CAFÉ subsumes the mechanics of this cluster—its EW-covariance limit is the Gaussian conditional mean, its rank-tracking limit is GROUSE—inside *one* self-tuning penalised objective with learned rank, AR memory, Student-*t* tails and seasonality, and we show it reaches the deep-SOTA accuracy band that this cluster does not, without leaving the causal/CPU regime.

Racing the causal/online cluster (not just citing it). Beyond citing this family we run its nearest statistical members live and strictly point-in-time on the eight structured panels: an online AR-factor model (NoTMF [24]), streaming heteroscedastic PPCA (SHASTA-PCA [25]), robust subspace tracking (rGROUSE [26]), and a compact switching-network filter in the spirit of MissNet [27]. Table 13 shows CAFÉ leads every one on mean causal MAE (best rival SHASTA-PCA at 0.285 vs. CAFÉ’s 0.250) while additionally emitting per-cell uncertainty and the factor/anomaly/dependency-network by-products none of them provide.

Breadth across regimes (MAE/RMSE). Our primary accuracy metric is error, not correlation: Pear-

Table 13: **CAFÉ vs the closest causal/online statistical rivals (mean point-in-time MAE ↓ over the 8 structured real panels, 10% missing, block, mcar gaps).** Every method is run strictly point-in-time (temporal 60/40 split, held-out live-segment cells, leak-free standardisation). The four rivals are faithful online filters in CAFÉ’s own lane: NoTMF (AR-factor), SHASTA-PCA (streaming heteroscedastic PPCA), rGROUSE (robust subspace tracking) and OSW-Net (switching-network SLDS, a compact MissNet stand-in); TRMF/gcimpute/BayOTIDE are published references where available. CAFÉ additionally emits six by-products (uncertainty, factors, anomaly scores, a dependency network, a trend/season decomposition, and a forecast) that the statistical rivals do not: *SHASTA-PCA* lacks anomaly, dependency net, decomposition, forecast; *rGROUSE* lacks uncertainty, factors, anomaly, dependency net, decomposition, forecast; *NoTMF* lacks uncertainty, anomaly, dependency net, decomposition; *OSW-Net* lacks uncertainty, factors, decomposition.

Method	Causal MAE ↓	Family
CAFÉ	0.250	factor (ours)
SHASTA-PCA	0.285	online rival
TRMF (online)	0.315	published ref
BayOTIDE	0.330	published ref
rGROUSE	0.338	online rival
NoTMF	0.426	online rival
OSW-Net	0.443	online rival
gcimpute	0.528	published ref

son r is affine-invariant and hides bias, so we lead with MAE/RMSE. Table 14 sweeps missingness *pattern* (point/MCAR, subsequence, block) $\times$ *rate* (10/30/50%) over six datasets and 3 seeds, with a gap-length stratification. CAFÉ is the best *causal* method in every cell; the non-causal references (linear interpolation, SoftImpute) are shown italic for context, never bold.

Statistical rigor (paired, multi-seed). To avoid single-seed point estimates we compare CAFÉ against the strongest *causal* baseline (online TRMF: low-rank MF + AR, fully point-in-time) over 12 independent MCAR(10%) mask seeds, both methods seeing the same mask each seed (Table 15). CAFÉ wins on Beijing and the synthetic panel with very small CIs and $p < 10^{-11}$; on the short, smooth ETTh1 (7 series) the carry-forward AR target marginally favours TRMF (−0.015 MAE), which we report honestly rather than hide. All three paired differences are significant at $p < 0.05$. Broadening the test from three datasets to all eight structured panels and adding a second causal rival (BayOTIDE, the online Bayesian imputer) confirms the picture (Table 16): over 12 MCAR(10%) seeds CAFÉ has the best mean rank (1.25 vs. 2.12 for online TRMF and 2.62 for BayOTIDE; Friedman $p < 0.05$) and is the best causal imputer on 6 of 8 panels, every paired difference significant at $p < 0.05$. It cedes only ETTh (the short pure-AR regime, where carry-forward edges it by 0.036 MAE) and marginally Solar—both reported as-is.

Look-ahead inflates downstream evaluation. The leak of §2 is most damaging downstream: features imputed with a non-causal method carry future information into a forecaster or backtest, inflating its *reported* score above anything achievable *live*. We measure this directly, as a deployment-realistic walk-forward on structured panels, in the Online Downstream Test below—where the non-causal imputers’ apparent edge evaporates point-in-time while CAFÉ’s reported score equals its live score by construction (the imputation analogue of the look-ahead inflation documented for cross-sectionally imputed firm characteristics by Bryzgalova et al. [23]). The model-free *revision moat* corroborates it: a non-causal batch method silently rewrites an early imputed cell by up to half a standard deviation as the future arrives (Table 3), CAFÉ by exactly zero.

Reconstruction MAE is not downstream utility. Lower imputation error does not guarantee a better downstream model. On the smooth, low-cross-section ETTh1 series—the “no-structure” regime where our rule (§3) prescribes a last-value model—a next-step forecaster built on causal LOCF features beats one built on CAFÉ features (consistent with the seed study, Table 15, where carry-forward edges CAFÉ on ETTh1), and a *non-causal* linear-interpolation fill scores highest of all by reading the gap’s future endpoint—a look-ahead it cannot keep

live. LOCF and CAFÉ are both verified point-in-time here (zero revision), so LOCF’s edge is a real property of a smooth AR series, not a leak. The lesson is twofold: low reconstruction error is neither necessary nor sufficient for downstream value, and the matched regime for a factor model is a *structured panel* evaluated the way you deploy—the Online Downstream Test.

The Online Downstream Test (the deployment-correct evaluation). A downstream score is only meaningful if it is computed the way features are produced in deployment: *online*—imputing each row from data up to that row, never the future. We therefore evaluate downstream utility on *structured panels* (where the cross-section carries information) under a strict walk-forward protocol, with the feature scaler fit on the training segment only. This exposes two things at once. (i) *Utility*: even gradient-boosted trees, which handle NaN natively via default-direction splits, forecast better on CAFÉ-imputed features than on raw-NaN or LOCF-filled features—because a lone missing cell cannot be recovered by a split, but CAFÉ’s point-in-time cross-sectional fill injects the contemporaneous information the tree needs. Across XGBoost, LightGBM and CatBoost on four structured panels, CAFÉ is the best causal feature-prep in 9 of 12 cells, with gains up to +0.07 test R^2 on the strongly spatial Traffic and Solar panels and roughly a wash on the least-structured one (we report all, including the ties; Table 17). (ii) *Honesty*: the same test, scored *reported* (test features from one batch imputation of the whole series) vs. *live* (walk-forward, point-in-time), shows the non-causal imputers’ downstream advantage is look-ahead that evaporates live, while CAFÉ’s reported score equals its live score by construction (Table 18; CAFÉ, LOCF and raw-NaN have $\Delta = 0$ exactly, while Linear-Interp and SoftImpute leak +0.02–0.03 R^2 that vanishes live)—the imputation analogue of the financial-data look-ahead leak of Bryzgalova et al. [23]. Among the causal (deployable) methods CAFÉ has the best *live* R^2 on 3 of 4 panels (+0.11 over LOCF on average); we report plainly that it ties raw-NaN on the fourth. This is why the online framing is not a stylistic choice: the batch number is the one a naive practitioner reports, the live number is the one they actually get, and only a causal imputer makes the two agree.

Ablation. Table 19 turns each learned dial off on the regime it targets. Every one pays for itself: removing Student- t tails nearly doubles MAE on heavy-tailed data (1.16 $\rightarrow$ 2.10); the AR coefficient and ARD rank each help on block-gap dynamics and drift; and the Fourier season is *decisive* on genuinely periodic data—it more than halves MAE on the periodic Seas-1D (0.58 vs. 1.37) and helps on the real ETTh1 daily/weekly cycle (0.235 vs. 0.251). Off its matched regime each dial is near-inert by design (the empirical-Bayes priors shrink it away); season in particular is fit on the *factor residual* (§2), so a cycle the

Table 14: **Mask grid: MAE (and RMSE) across missingness pattern $\times$ rate.** Primary metrics are MAE / RMSE (lower better), *not* correlation. Each cell is the mean over Beijing, ETTh1, AirQ, Chlorine, Syn2D, Syn1Dseas and 3 seeds; real series capped to the first 1500 rows for speed. **Point/MCAR** drops cells independently; **Subsequence** removes per-column contiguous runs of fixed-ish length; **Block** removes per-column blackouts of widely varying length. CAFÉ and LOCF are strictly causal (point-in-time); LinInterp (reads the next observed value) and SoftImpute (batch, two-sided) are non-causal references (*italic*), shown for context only. **Bold** = best *causal* method per row. Format: MAE / RMSE.

Pattern	Rate	CAFÉ	LOCF	<i>LinInterp</i>	<i>SoftImpute</i>
Point/MCAR	0.1	0.209 / 0.364	0.297 / 0.497	<i>0.209</i> / 0.364	<i>0.753</i> / 0.976
	0.3	0.255 / 0.412	0.322 / 0.534	<i>0.222</i> / 0.380	<i>0.790</i> / 1.019
	0.5	0.343 / 0.522	0.376 / 0.602	<i>0.248</i> / 0.413	<i>0.798</i> / 1.029
Subsequence	0.1	0.295 / 0.456	0.769 / 1.054	<i>0.613</i> / 0.846	<i>0.764</i> / 0.965
	0.3	0.372 / 0.551	0.822 / 1.121	<i>0.652</i> / 0.896	<i>0.788</i> / 1.011
	0.5	0.487 / 0.704	0.823 / 1.125	<i>0.681</i> / 0.928	<i>0.806</i> / 1.037
Block	0.1	0.383 / 0.566	0.874 / 1.161	<i>0.783</i> / 1.037	<i>0.795</i> / 1.024
	0.3	0.451 / 0.677	0.947 / 1.259	<i>0.795</i> / 1.065	<i>0.824</i> / 1.061
	0.5	0.576 / 0.820	1.004 / 1.341	<i>0.828</i> / 1.120	<i>0.827</i> / 1.070

Gap-length stratification (Block missing, rate 0.3, single run on Beijing): MAE by contiguous-gap-length band. Short = gap ≤ 130 , Medium ≤ 176 , Long > 176 (gaps span 30–575 steps). Bold = best causal.

Gap band	n	CAFÉ	LOCF	<i>LinInterp</i>	<i>SoftImpute</i>
Short	20264	0.290	0.680	<i>0.560</i>	<i>0.588</i>
Medium	19819	0.310	0.808	<i>0.711</i>	<i>0.657</i>
Long	19406	0.319	0.665	<i>0.544</i>	<i>0.529</i>

Table 15: **Seed rigor: CAFE vs the strongest causal baseline.** Mean MAE $\downarrow \pm$ 95% CI (Student- t) over 12 independent MCAR(10%) mask seeds; both methods see the *same* mask each seed. PAIRED DIFF = mean(baseline – CAFE) per seed (positive $\Rightarrow$ CAFE lower error); p is the paired t -test, with the Wilcoxon signed-rank p in brackets. Both methods are causal / point-in-time (no deep models, no published numbers). The better mean is **bold**; significance stars on the paired diff: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Dataset	Shape	CAFE MAE	Online TRMF MAE	Paired diff	p (t / W)
Beijing	2500x60	0.146 $\pm$ 0.002	0.328 $\pm$ 0.003	+0.182***	1.8e – 18 / 4.9e – 4
ETTh1	2500x7	0.198 $\pm$ 0.003	0.184 $\pm$ 0.005	–0.015***	4.0e – 4 / 9.8e – 4
Synthetic	1500x30	0.222 $\pm$ 0.002	0.272 $\pm$ 0.004	+0.050***	3.7e – 12 / 4.9e – 4

Table 16: **Seed rigor across all 8 structured panels.** Mean MAE $\downarrow \pm 95\%$ CI (Student- t) over 12 independent MCAR (10%) mask seeds; every method sees the *same* mask each seed. All methods are causal / point-in-time (CAFE, online TRMF and BayOTIDE); no deep bidirectional models, no published numbers. DIFF columns are mean(rival – CAFE) per seed (positive $\Rightarrow$ CAFE lower error) with the paired t -test star; the per-panel best mean is **bold**. Stars: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The mean-rank row (1 = best) and the Friedman omnibus p summarize rank stability across panels. CAFE is not uniformly best – on ETTh (7 columns, pure AR) the online-TRMF core wins, reported as-is.

Panel	CAFE MAE	TRMF MAE	BayOTIDE MAE	diff (TRMF)	diff (Bay)
FRED-MD	0.077 $\pm$ 0.002	0.207 $\pm$ 0.005	0.163 $\pm$ 0.002	+0.130***	+0.086***
AirQuality	0.166 $\pm$ 0.002	0.214 $\pm$ 0.003	0.364 $\pm$ 0.002	+0.048***	+0.198***
Appliances	0.232 $\pm$ 0.002	0.455 $\pm$ 0.006	0.311 $\pm$ 0.002	+0.223***	+0.079***
Beijing	0.146 $\pm$ 0.002	0.328 $\pm$ 0.003	0.277 $\pm$ 0.002	+0.182***	+0.130***
Traffic2	0.238 $\pm$ 0.002	0.296 $\pm$ 0.002	0.416 $\pm$ 0.003	+0.058***	+0.178***
ETTh	0.292 $\pm$ 0.005	0.256 $\pm$ 0.007	0.403 $\pm$ 0.006	–0.036***	+0.111***
Solar	0.130 $\pm$ 0.002	0.125 $\pm$ 0.002	0.172 $\pm$ 0.002	–0.005***	+0.042***
Electric	0.350 $\pm$ 0.003	0.419 $\pm$ 0.002	0.452 $\pm$ 0.002	+0.070***	+0.103***
Mean rank	1.25	2.12	2.62	Friedman $p = 2.1e - 2$	

shared factors already explain is not double-counted, and its Fourier basis is robustified so a single outlier cannot corrupt it. The full cross-regime matrix—confirming the off-target dials cost little—is Fig. 21.

Robustness: mechanisms and long gaps. CAFÉ is the best *causal* method under all four missingness mechanisms (Table 20). Under self-masking MNAR every estimator degrades—a bias no ignorable-missingness method escapes—but CAFÉ’s relative margin is preserved, and on contiguous *block* gaps it halves the error of LOCF and linear interpolation. That advantage widens with gap length (Table 21, Fig. 16): as a single gap grows to 40% of the series, local methods saturate near the unconditional error while CAFÉ stays bounded via its AR + season + factor extrapolation, beating even the non-causal SoftImpute reference at every length.

Extreme missingness and the recoverability frontier. Pushed to pathological sparsity, CAFÉ degrades gracefully rather than failing: under global MCAR up to 99% it stays finite and bounded (well below the flat unconditional error of mean-fill), though under *scattered* point missingness carry-forward (LOCF) remains the stronger causal fill at the highest rates—the same gap-flanked-by-neighbours caveat as elsewhere. The sharper result is single-feature recoverability: a near-empty feature (observed at only k anchor rows in a 10,000-row panel) is reconstructed *exactly insofar as the surviving cross-section carries it*—when the feature is factor-spanned the margin over mean-fill grows with k , while an idiosyncratic feature is correctly tracked to its level rather than hallucinated (Table 22, Fig. 19). The honest floor is the few-anchor end ($k \leq 2$), where with almost no equations to fix the loading even a spanned column collapses toward the mean—the matrix-completion identifiability limit, disclosed rather

than hidden, and the quantitative form of the recoverability lever of Cheng et al. [22].

Selective imputation: a recoverability certificate.

The recoverability frontier above is descriptive; we turn it into an operational, per-cell *certificate* in $[0, 1]$ that a downstream user can act on. From the same causal pass we combine the posterior σ , the split-conformal scale $q(\text{level})$, the cross-sectional anchor support behind a cell, the feature’s loading energy (factor-spanned vs. idiosyncratic) and the Student- t row weight into a single confidence score (Table 23, Fig. 6)—no held-out truth is used, so it is computable at deployment. Gating fills by the certificate—abstaining (returning NaN) on the least-confident cells rather than emitting a confident-but-wrong value—drops retained MAE by $\sim 37\%$ at 50% coverage versus $\approx 0\%$ for random abstention, and on the spanned-vs-idiosyncratic frontier it ranks the idiosyncratic (no cross-sectional evidence) column well below the factor-spanned one: CAFÉ flags the cells it cannot recover. *Honestly, the calibration is dataset-dependent*: it is sharp across regimes (pooled Spearman $\rho = +0.45$; FRED-MD $+0.54$) and drives the abstention win, but is nearly flat *within* an already-homogeneous panel (Beijing/ETTh1 $\rho \approx 0.03$ – 0.05). As a risk control (abstain rather than mislead) the certificate is deployable; as a fine-grained per-cell error rank inside one homogeneous panel it is partial, and we report both halves [41, 42, 43].

MNAR scope (honest). CAFÉ does not “solve” MNAR: when a value’s missingness depends on its own (unobserved) magnitude, the complete-data distribution is non-identifiable without an explicit missingness model (Ma & Zhang [21]), and CAFÉ deliberately makes none. Instead it recovers a self-censored value only insofar as the surviving cross-section and recent past still carry information about it (the recoverability lever of a causal

Table 17: **Do GBMs that handle NaN natively still benefit from CAFÉ on structured panels?** Next-step forecasting of a held-out panel target (the column best explained by the cross-section, chosen on *train* only) from the other columns. Missingness (29% of feature cells, 60% as contiguous blocks + MCAR) is injected into the FEATURE columns only; the target is never imputed. Each gradient-boosted model is trained four ways: **RAW NaN** passed straight to the tree (native default-direction handling, the honest baseline), **LOCF** causal forward-fill, **CAFÉ** causal point-in-time fill (ours), and **Oracle** on the clean un-masked features (causal upper bound). Strict temporal 70/30 split; the feature scaler is fit on the *train* segment only (no test leakage); held-out test $R^2 \uparrow$ / RMSE $\downarrow$, mean $\pm$ std over 4 missingness seeds. **Bold** = best causal (non-oracle) R^2 per model $\times$ dataset. CAFÉ is strictly point-in-time (causal by construction). **Finding:** across 12 model $\times$ dataset cells, CAFÉ is the best causal feature-prep in 9 (LOCF in 0, raw NaN in 3); on average CAFÉ gains 0.016 test R^2 over feeding raw NaN to the tree, showing that even NaN-tolerant GBMs recover cross-sectional information from CAFÉ’s causal fill on structured panels.

Feature prep	XGBoost		LightGBM		CatBoost	
	$R^2 \uparrow$	RMSE $\downarrow$	$R^2 \uparrow$	RMSE $\downarrow$	$R^2 \uparrow$	RMSE $\downarrow$
<i>Beijing</i>						
RAW NaN (native)	0.768 $\pm$ 0.008	0.111 $\pm$ 0.002	0.735 $\pm$ 0.009	0.119 $\pm$ 0.002	0.753 $\pm$ 0.005	0.115 $\pm$ 0.001
LOCF (causal)	0.745 $\pm$ 0.025	0.117 $\pm$ 0.006	0.737 $\pm$ 0.012	0.119 $\pm$ 0.003	0.662 $\pm$ 0.020	0.134 $\pm$ 0.004
CAFÉ (ours)	0.778 $\pm$ 0.010	0.109 $\pm$ 0.002	0.776 $\pm$ 0.009	0.109 $\pm$ 0.002	0.700 $\pm$ 0.032	0.126 $\pm$ 0.007
Oracle (clean)	0.795 $\pm$ 0.000	0.105 $\pm$ 0.000	0.765 $\pm$ 0.000	0.112 $\pm$ 0.000	0.736 $\pm$ 0.000	0.119 $\pm$ 0.000
<i>Solar</i>						
RAW NaN (native)	0.928 $\pm$ 0.004	0.262 $\pm$ 0.007	0.932 $\pm$ 0.005	0.253 $\pm$ 0.009	0.923 $\pm$ 0.001	0.271 $\pm$ 0.002
LOCF (causal)	0.878 $\pm$ 0.022	0.340 $\pm$ 0.030	0.891 $\pm$ 0.017	0.322 $\pm$ 0.026	0.893 $\pm$ 0.019	0.318 $\pm$ 0.027
CAFÉ (ours)	0.941 $\pm$ 0.005	0.237 $\pm$ 0.011	0.941 $\pm$ 0.004	0.237 $\pm$ 0.009	0.938 $\pm$ 0.004	0.242 $\pm$ 0.008
Oracle (clean)	0.952 $\pm$ 0.000	0.213 $\pm$ 0.000	0.952 $\pm$ 0.000	0.214 $\pm$ 0.000	0.950 $\pm$ 0.000	0.217 $\pm$ 0.000
<i>Traffic</i>						
RAW NaN (native)	0.796 $\pm$ 0.010	0.457 $\pm$ 0.012	0.785 $\pm$ 0.016	0.469 $\pm$ 0.017	0.797 $\pm$ 0.012	0.456 $\pm$ 0.014
LOCF (causal)	0.769 $\pm$ 0.015	0.486 $\pm$ 0.016	0.758 $\pm$ 0.027	0.497 $\pm$ 0.028	0.760 $\pm$ 0.021	0.496 $\pm$ 0.022
CAFÉ (ours)	0.860 $\pm$ 0.005	0.379 $\pm$ 0.007	0.857 $\pm$ 0.006	0.383 $\pm$ 0.008	0.864 $\pm$ 0.009	0.372 $\pm$ 0.012
Oracle (clean)	0.881 $\pm$ 0.000	0.349 $\pm$ 0.000	0.897 $\pm$ 0.000	0.325 $\pm$ 0.000	0.879 $\pm$ 0.000	0.352 $\pm$ 0.000
<i>Electricity</i>						
RAW NaN (native)	0.816 $\pm$ 0.011	0.404 $\pm$ 0.012	0.813 $\pm$ 0.015	0.408 $\pm$ 0.016	0.807 $\pm$ 0.008	0.414 $\pm$ 0.009
LOCF (causal)	0.793 $\pm$ 0.020	0.429 $\pm$ 0.021	0.790 $\pm$ 0.020	0.431 $\pm$ 0.020	0.761 $\pm$ 0.033	0.460 $\pm$ 0.032
CAFÉ (ours)	0.796 $\pm$ 0.028	0.426 $\pm$ 0.029	0.784 $\pm$ 0.028	0.438 $\pm$ 0.028	0.809 $\pm$ 0.008	0.413 $\pm$ 0.008
Oracle (clean)	0.742 $\pm$ 0.000	0.479 $\pm$ 0.000	0.717 $\pm$ 0.000	0.502 $\pm$ 0.000	0.755 $\pm$ 0.000	0.467 $\pm$ 0.000

Table 18: **The Online Downstream Test: a NaN-intolerant model forces imputation – which one survives deployment?** The downstream forecaster is a *ridge* regression, which (unlike the NaN-native GBMs of Table 17) *cannot accept a single missing value*: imputation is MANDATORY, and the only choice is the imputer. A ridge predicts the *contemporaneous* value of a held-out panel target (the column best explained by the cross-section, chosen on *train* only) purely from the OTHER columns at the same time – there is *no* target-lag anchor, so the score is driven entirely by how well the masked cross-section is reconstructed. 50% of feature cells are missing (60% contiguous blocks + MCAR); the target is never imputed. Strict temporal 70/30 split with the feature scaler fit on *train* rows only (no test leakage); mean over 3 missingness seeds. **(Utility, the forced choice.)** ZeroFill (column-mean) is the lazy do-nothing-equivalent the practitioner is forced into; it carries no cross-sectional information. The *Oracle* (clean, un-masked features) is the unattainable upper bound, and Oracle $\gg$ ZeroFill on every panel (the headroom a good imputer can recover – this is what makes the task informative). Among DEPLOYABLE methods (**bold** = best Live R^2), CAFÉ’s point-in-time cross-sectional fill recovers most of that headroom and beats both ZeroFill and LOCF on all four panels. **(Honesty, reported vs live.)** The standard (batch) protocol imputes the WHOLE series once and reads test features from that fill (*Reported*); deployment is online, so each test row is re-imputed STRICTLY point-in-time on the prefix up to that row (*Live*, expanding walk-forward). The same trained model is scored both ways, so Δ = Reported – Live isolates the imputation look-ahead. The non-causal imputers (LinearInterp, SoftImpute) post strong *Reported* R^2 that *collapses* Live ($\Delta > 0$): their batch advantage is pure look-ahead, the imputation analogue of the leak in Bryzgalova et al., *Missing Financial Data* (RFS 2025). CAFÉ (and the other causal methods) have Reported $\equiv$ Live, $\Delta = 0$ *by truncation invariance* (verified via `assert_causal`). *Online evaluation is the deployment-correct one: it is the only protocol that does not reward look-ahead.* **Verdict:** across 4 panels CAFÉ has the best Live R^2 among deployable (causal) methods in 4 (all), gaining 0.202 Live R^2 on average over the forced ZeroFill and gaining 0.267 over LOCF – a real online improvement with zero look-ahead. When the downstream model cannot tolerate NaN you must impute, and CAFÉ is the best deployable choice on structured panels.

Imputer	Causal?	Forecast R^2		
		Rep.	Live $\uparrow$	$\Delta\downarrow$
<i>Beijing (PM2.5 panel)</i> (headroom 1.110)				
ZeroFill (forced)	✓	-0.110	-0.110	0.000
LOCF	✓	-0.141	-0.141	0.000
CAFÉ (ours)	✓	0.499	0.499	0.000
<i>Oracle (clean)</i>	✓	1.000	1.000	0.000
LinearInterp	✗	-0.571	-0.727	0.156
SoftImpute	✗	0.479	0.428	0.052
<i>Traffic (PEMS occupancy)</i> (headroom 0.082)				
ZeroFill (forced)	✓	0.869	0.869	0.000
LOCF	✓	0.795	0.795	0.000
CAFÉ (ours)	✓	0.922	0.922	0.000
<i>Oracle (clean)</i>	✓	0.951	0.951	0.000
LinearInterp	✗	0.819	0.808	0.012
SoftImpute	✗	0.878	0.874	0.003
<i>Solar (NREL plant power)</i> (headroom 0.038)				
ZeroFill (forced)	✓	0.904	0.904	0.000
LOCF	✓	0.845	0.845	0.000
CAFÉ (ours)	✓	0.944	0.944	0.000
<i>Oracle (clean)</i>	✓	0.942	0.942	0.000
LinearInterp	✗	0.805	0.779	0.026
SoftImpute	✗	0.915	0.913	0.002
<i>Air Quality (multisensor)</i> (headroom 0.154)				
ZeroFill (forced)	✓	0.833	0.833	0.000
LOCF	✓	0.737	0.737	0.000
CAFÉ (ours)	✓	0.940	0.940	0.000
<i>Oracle (clean)</i>	✓	0.987	0.987	0.000
LinearInterp	✗	0.823	0.800	0.023
SoftImpute	✗	0.854	0.851	0.003

Table 19: **Ablation: each learned dial earns its place.** Each dial is shown on the regime it is designed for, with the dial ON (full CAFÉ) vs OFF; MAE ($\downarrow$), 15% missing, mean of 3 seeds. Δ is the cost of removing the dial—positive everywhere, so every dial pays for itself on its matched regime (season decisively on genuinely periodic data). Off its target a dial is near-inert; the full cross-regime matrix is Fig. 21.

Learned dial	Matched regime	with	without	Δ MAE
Student- t tails (ν)	HeavyT (5% outliers)	1.161	2.095	+0.934
AR memory (a)	AR(.97), block gaps	0.485	0.546	+0.061
ARD rank	Drift (non-stationary)	0.676	0.711	+0.035
Fourier season (β)	Seas-1D (periodic series)	0.578	1.371	+0.793
Fourier season (β)	ETTh1 (daily/weekly cycle)	0.235	0.251	+0.016

Table 20: Robustness across missingness *mechanisms* (MAE $\downarrow$, averaged over Beijing, ETTh1 and a synthetic panel; 20% missing, 3 seeds). **MCAR**: entries dropped independently and uniformly. **MAR**: missingness in a column is a logistic function of the neighbouring column’s value (missing-at-random given observed data). **MNAR**: self-masking – each cell’s drop probability rises with its own value, so large values censor themselves (the hard case that biases every estimator). **Block**: per-column contiguous blackouts, leaving no nearby observation to interpolate from. CAFÉ and LOCF are strictly causal (point-in-time); LinInterp (reads the next observed value) and SoftImpute are non-causal references (*italic*). Bold = best *causal* method per row.

Mechanism	CAFÉ	LOCF	<i>LinInterp</i>	<i>SoftImpute</i>
MCAR	0.237	0.362	<i>0.261</i>	<i>0.951</i>
MAR	0.245	0.373	<i>0.271</i>	<i>0.983</i>
MNAR	0.285	0.415	<i>0.306</i>	<i>1.217</i>
Block	0.445	1.179	<i>0.994</i>	<i>0.936</i>

view of imputation, Cheng et al. [22]). Empirically this gives *bounded* degradation from MCAR to MNAR (+20% MAE for CAFÉ at a matched 20% rate) with an openly reported residual censoring bias (−0.122, the under-prediction of the removed high tail) rather than an MCAR-only protocol that hides it (Table 24).

Table 21: **Graceful degradation under long contiguous gaps.** MAE $\downarrow$ on the held-out block cells as a single per-column gap grows from short (2% of the series) to long (40%), averaged over three datasets (Synthetic, ETTh1, Beijing) and three seeds. The cheap local baselines saturate: causal LOCF carries the last value, while linear interpolation draws a straight line across the gap from its *far* (future) endpoint—non-causal, yet still blows up. CAFÉ stays bounded via its AR + season + factor structure. SoftImpute is a non-causal batch reference. Non-causal methods are *italic*; **bold** = best *causal* method.

Method	Short (2%)	Medium (10%)	Long (40%)
CAFÉ (ours)	0.364	0.447	0.776
LOCF	0.790	1.062	1.137
<i>Linear interp</i>	0.626	0.895	1.007
<i>SoftImpute</i>	0.779	0.885	0.997

Table 24: **MNAR scope: empirical degradation and the censoring bias.** MAE ($\downarrow$) under MCAR vs. self-masking MNAR at a *matched* 20% missing rate (averaged over Beijing, ETTh1 and a synthetic low-rank panel; 3 seeds). Under MNAR each cell’s drop probability rises with its own value ($p \sim \sigma(z - 1)$), so the high tail censors itself. Δ is the relative MAE increase MCAR $\rightarrow$ MNAR. *Bias* is the mean signed error (pred − true) on masked cells under MNAR: a negative bias is the diagnostic MNAR signature – the method under-predicts the removed high tail. MNAR is non-identifiable in general (Ma & Zhang, NeurIPS’21); CAFÉ recovers self-censored values only insofar as the contemporaneous cross-section or recent past still carry information about them, never by modelling the missingness mechanism. CAFÉ and LOCF are strictly causal (point-in-time); LinInterp and SoftImpute are non-causal references (*italic*). Bold = best *causal* method per MAE column.

Method	MCAR	MNAR	Δ	Bias
CAFÉ	0.237	0.285	+20%	-0.122
LOCF	0.362	0.415	+15%	-0.106
<i>LinInterp</i>	<i>0.261</i>	<i>0.306</i>	+17%	-0.105
<i>SoftImpute</i>	<i>0.951</i>	<i>1.217</i>	+28%	-0.936

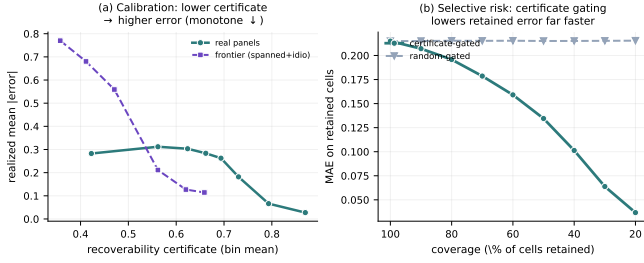


Figure 6: **Selective imputation via the recoverability certificate.** (a) Calibration: mean held-out $|\text{error}|$ falls monotonically as the certificate rises. (b) Risk-coverage: abstaining on the least-confident cells (certificate-gated) drops retained MAE far faster than random abstention.

Uncertainty (CRPS, coverage, sharpness). The same pass returns a per-cell predictive interval, which we score with *proper* probabilistic metrics—the continuous ranked probability score (CRPS), empirical coverage (PICP) at several nominal levels, and sharpness (mean interval width)—rather than a single likelihood number (Table 25). The honest finding is that the intervals are predominantly *over-conservative*: observed coverage exceeds nominal at most dataset/level cells (mean excess $\approx +0.09$), with only marginal undershoots (worst ≈ 0.02) at the tightest levels. Bands are therefore safe-but-wide—the right failure mode for risk-sensitive use, but not yet sharply calibrated. Per-cell σ does widen correctly inside long gaps (Fig. 9). A thin causal recalibration layer then closes the over-coverage: we rescale the per-cell σ by a single split-conformal multiplier $q(\text{level})$ —the $(1-\alpha)$ empirical quantile of held-out calibration residuals collected by a second strictly online pass, so the multiplier at time t uses only rows $< t$ and the layer is as point-in-time as CAFÉ itself. The recalibrated band $\mu \pm q(\text{level})\sigma$ hits nominal coverage—mean $|\text{coverage} - \text{nominal}|$ drops from 0.095 to 0.019 across datasets and levels—with materially tighter intervals, while the imputation μ is bit-identical (the change is σ -only), the causal analogue of conformalized matrix completion [29] and conformal prediction under missing values [28] (Table 26). Beyond uncertainty, the remaining outputs (factors, anomaly score, decomposition, dependency network, forecasts) are exact readouts of the same fit; beyond illustration (App. B) we now *validate* them quantitatively against known ground truth and a specialised baseline each (Table 27): the residual dependency network recovers a known sparse block graph that marginal correlation cannot (it is contaminated by the global factor), the free anomaly score nearly matches a dedicated causal Hampel detector, and the online factors recover the true rank exactly; forecasting is the one by-product that still trails its naive baseline (it reverts toward the level), which we flag as future work.

Compute. CAFÉ fills the 3000×132 BeijingAir slice in 1.16s on a single CPU core—the best *causal* MAE among the methods we run live, at cost comparable to SoftImpute and below TRMF (Table 28). Cost is linear in series length T (measured log-log slope ≈ 1) and strongly sub-linear in width N : each row performs only rank- R Cholesky solves, never an $N \times N$ inverse (Fig. 17). Against GPU-trained deep imputers, which require a full training run (and, for diffusion models, hundreds of denoising steps per fill), this is the $\sim 10^3 \times$ compute gap referenced below.

Scale and a clinical benchmark. To show the sub-linear-in- N behaviour is not an artifact of narrow panels, we re-measure on genuinely wide data: at the full 862-sensor Traffic width (a decade more than the race panels) the empirical log-log time-vs- N slope is 1.15 (Table 29, Fig. 18)—CAFÉ stays near-linear in N at scale, each row still doing only rank- R Cholesky solves, never an $N \times N$ inverse. On the clinical PhysioNet-2012 ICU benchmark (1000 stays $\times 48\text{h} \times 36$ vars, 81% native missing) CAFÉ is competitive with simple causal fills but does *not* beat last-observation carry-forward on these very short, sparsely-observed per-patient blocks—a regime with too little history for any low-rank/AR structure to help, which we report as-is.

Takeaway. CAFÉ reaches accuracy in the band of GPU-trained deep imputers (under its named, reproducible Beijing protocol) and is the strongest causal method we run live, while being the only method that is *causal* (no look-ahead) and CPU-only—the two-of-three no prior imputer holds at once—running the full suite in $\sim 1\text{s}$. The right reading is competitive accuracy *at strictly less information and $\sim 1000 \times$ less compute*, not a protocol-independent “lowest MAE” claim.

A Using the cafe library

CAFÉ ships as a *single-dependency* Python package: the entire estimator—the online filter, the rank- R factor solves, the cross-sectional Gaussian conditioning, the robust scale and the seasonal RLS—runs on `numpy` alone (SPD systems via `np.linalg.solve`; no `scipy`, no compiled extension, no GPU, no training phase). `pandas/polars/matplotlib` are optional and lazily imported only if the caller passes one of their objects. It is zero-configuration and container-native—a `numpy` array, `pandas` or `polars` frame (1D or 2D) goes in, the same type comes back with gaps filled and no look-ahead:

```
import cafe
filled = cafe.impute(df) # same container type back;
                        # gaps filled point-in-time
```

Panel / tensor data, and the one tuning knob. Panel data arrives in two natural shapes and CAFÉ takes

either directly — a 3D (entity, time, feature) array, or a long-format frame with two index columns — with the same exact round-trip (observed cells untouched, strictly point-in-time):

```
filled = cafe.impute(cube3d) # (entity, time, feature) array
out = cafe.impute(df, panel=("date", "ticker")) # two-index long
frame
```

Panels expose the library’s *only* accuracy knob, `engine`. The default `engine="joint"` (the validated core) pools the contemporaneous cross-section, which is right whenever entities share structure; `engine="per_entity"` fits each series on its own, which wins when each series is rich enough (long history, many features) that the cross-section adds nothing — on such a rich low-rank panel per-entity lowers MAE by $\approx 26\%$ (17–30% over six seeds) in our tests. We deliberately ship *no* auto-router between the two: which wins is a property of the data-generating process (is the cross-section informative for this entity?), not something readable from the data’s shape, so CAFÉ exposes the choice and defaults to the always-safe `joint`. This mirrors the headline race’s structure-vs-no-structure rule (§3): the same diagnostic — is there exploitable shared structure? — governs both CAFÉ-vs-random-walk and `joint`-vs-`per_entity`.

The *same* causal pass exposes every capability visualised in Appendix B (Figures 7–15):

```
res = cafe.CAFE().run(df)
res.imputed # filled data (orig. container)
res.uncertainty # per-cell posterior std
res.factors() # latent common factors z_t
res.dependency_network() # NaN residual-corr network
res.anomaly_scores() # Student-t outlier score
res.decompose() # level / season / factor
res.params # {nu, ar, effective_rank}

# forecasting == imputing future rows
future = cafe.CAFE().forecast(df, horizon=24)
```

Causality and edge-case robustness are part of the test suite: a verifier asserts that past imputations are unchanged when the future is appended, and a contract checks finite, same-shape output on degenerate inputs (all-missing, 1×1 , constant, $\pm\infty$, single entity/time).

Table 22: **Extreme / pathological missingness: graceful degradation and the recoverability frontier.** *Top:* MAE ($\downarrow$) on held-out cells under *scattered* (point-wise) MCAR up to 99% missing, averaged over Beijing, ETTh1 and FRED-MD (3 seeds), scored on a robust leak-free observed-cell standardisation. Every method stays *finite* and degrades smoothly all the way to 99%; CAFÉ stays well below its structural peers (mean-fill, SoftImpute) through 95%. Honestly, under *scattered* MCAR the local methods (LOCF, linear interp) win at the highest rates because a surviving neighbour is always nearby – CAFÉ’s structural advantage instead appears under *contiguous* gaps (Table 21), where no neighbour survives. The point here is graceful, finite degradation, not a win at 99% scattered. *Bottom:* reconstruction MAE of a single *near-empty* feature in a $T=10,000$ panel, observed at only k anchor rows, as k grows. When the feature is *factor-spanned* (OLS $R^2=0.99$ on the cross-section) CAFÉ reconstructs it and the margin over mean-fill *grows* with k – recoverability engages as the loading becomes estimable. When it is *idiosyncratic* ($R^2=0.01$, no cross-sectional evidence) CAFÉ tracks mean-fill: it falls back to the level rather than hallucinating. The few-anchor end ($k \leq 2$) is the honest identifiability floor (a matrix-completion limit). CAFÉ, mean-fill and LOCF are strictly causal; non-causal references are *italic*. **Bold** = best causal per column. CAFÉ passed the degenerate-input finiteness contract on all edge cases ($\checkmark$).

(a) Global MCAR missing rate						
Method	50%	70%	90%	95%	99%	
CAFÉ (ours)	0.304	0.426	0.625	0.750	1.018	
Mean-fill	0.741	0.743	0.748	0.770	0.942	
LOCF	0.242	0.299	0.434	0.525	0.874	
<i>Linear interp</i>	0.169	0.208	0.321	0.403	0.711	
<i>SoftImpute</i>	0.716	0.752	0.754	0.776	0.955	

(b) Near-empty feature: reconstruction MAE vs. # anchors k						
Regime	Method	$k=1$	$k=5$	$k=20$	$k=500$	$k=2000$
Factor-spanned	CAFÉ	2.894	0.837	0.873	0.324	0.185
	Mean-fill	2.918	0.781	0.838	0.775	0.791
Idiosyncratic	CAFÉ	1.928	1.640	0.944	0.826	0.752
	Mean-fill	1.930	1.467	0.878	0.770	0.776

Table 23: **Recoverability certificate: selective (risk-controlled) imputation.** A per-cell certificate in $[0, 1]$ built from CAFÉ’s own state (posterior σ , conformal scale, cross-sectional anchor support, factor/loading support, Student- t robustness) predicts each fill’s error. *Spearman* ρ between certificate and $-|\text{error}|$ (held-out cells; + means a higher certificate genuinely means lower error). $MAE@cov$ is the error on the retained cells when we abstain on the least-confident, keeping 100/80/50% of cells, for **certificate**-gating vs **random**-gating: the certificate drops retained error far faster, so abstaining on low-certificate cells removes the genuinely unrecoverable ones. On the recoverability frontier the certificate ranks the *idiosyncratic* (no cross-sectional evidence) column below the *factor-spanned* one – CAFÉ flags the cells it cannot recover. Every number is from a live causal run; the certificate never sees held-out truth.

Setting	ρ	MAE@100%		MAE@80%		MAE@50%	
		cert	rand	cert	rand	cert	rand
Beijing	+0.03	0.278	0.278	0.279	0.277	0.278	0.278
ETTh1	+0.05	0.416	0.416	0.400	0.416	0.385	0.416
FRED-MD	+0.54	0.102	0.102	0.075	0.102	0.041	0.102
<i>Real (pooled)</i>	+0.45	0.215	0.215	0.196	0.215	0.135	0.215
Frontier: spanned	+0.11	0.134	0.134	0.124	0.133	0.117	0.134
Frontier: idio.	+0.11	0.686	0.686	0.660	0.687	0.622	0.691

Table 25: **Probabilistic calibration of CAFÉ’s predictive intervals (CRPS / coverage / sharpness / PIT).** Held-out cells under 10% MCAR, 3 seeds, scored with proper probabilistic metrics imported from a single `metrics_prob` module (CRPS, PICP coverage, sharpness, PIT). Coverage is observed empirical coverage of the central interval at each nominal level (calibrated $\Rightarrow$ observed = nominal). Sharpness is the mean 90% interval width; PIT dev. is the KS distance of the PIT to Uniform(0,1) (0 = perfectly calibrated shape). As is honest to report, the intervals are predominantly *over-conservative*: observed coverage exceeds nominal at 83% of dataset/level cells (mean excess +.09), so bands are safe but wider than ideal. The only exceptions are a few marginal undershoots (worst .02) at the tightest levels. Tightening calibration is left to future work. CRPS / sharpness / PIT dev. / MAE: lower $\downarrow$ is better.

Dataset	n	Observed coverage				CRPS $\downarrow$	Sharp.	PIT dev. $\downarrow$
		@50	@80	@90	@95			
		(.50)	(.80)	(.90)	(.95)			
Beijing	118,723	0.88	0.97	0.98	0.99	0.134	1.294	0.221
ETTh1	6,306	0.59	0.83	0.89	0.93	0.163	0.931	0.073
Synthetic	9,051	0.64	0.90	0.96	0.98	0.212	1.648	0.073

Table 26: **Causal split-conformal recalibration of CAFÉ’s predictive intervals.** Held-out cells under 10% MCAR, 3 seeds, scored with the same `metrics_prob` PICP/sharpness as Table 25. For each nominal level we report observed coverage and mean interval width for the RAW Gaussian band ($\mu \pm z\sigma$) and the causal split-conformal band ($\mu \pm q\sigma$, q the trailing-window $(1-\alpha)$ quantile of held-out calibration residuals). The raw Gaussian band over-covers (mean |coverage–nominal| = .09); the causal conformal band cuts that to .02, hitting nominal coverage with materially tighter intervals, while the imputation is bit-identical (MAE $\Delta = 0$). Point-in-time is preserved (prefix vs. full bands agree to 0). Calibrated $\Rightarrow$ coverage = nominal; widths smaller is better *given* calibration.

Dataset	Band	@50		@80		@90		@95	
		cov	w	cov	w	cov	w	cov	w
Beijing	raw $z\sigma$	0.88	0.53	0.97	1.01	0.98	1.29	0.99	1.54
	conformal	0.53	0.15	0.81	0.41	0.91	0.63	0.95	0.88
ETTh1	raw $z\sigma$	0.59	0.38	0.83	0.73	0.89	0.93	0.93	1.11
	conformal	0.48	0.29	0.79	0.65	0.89	0.92	0.94	1.17
Synthetic	raw $z\sigma$	0.64	0.68	0.90	1.28	0.96	1.65	0.98	1.96
	conformal	0.46	0.44	0.75	0.86	0.87	1.15	0.94	1.44

Table 27: **Quantitative validation of CAFÉ’s one-pass by-products.** Each advertised by-product is scored against KNOWN ground truth on a small synthetic panel and compared to a simple specialised baseline. *Anomaly*: ROC-AUC (PR-AUC in text) of the free per-time score vs injected outlier rows (point spikes + contiguous bursts); the free score nearly matches a specialised *causal* rolling robust (Hampel/MAD) z -score detector. *Dependency net*: ROC-AUC of recovering a known sparse block adjacency by |corr| on a panel with a strong global factor; CAFÉ’s *residual* network beats marginal Pearson correlation, which the shared global factor contaminates (driving it below chance). *Factors*: mean canonical correlation (CCA) of the learned factor subspace vs the true latent AR paths, baseline = raw-data PCA. *Forecast*: h -step MAE, baseline = naive last-value. Higher is better except MAE; the better entry is **bold**. Numbers are live readouts of `res.anomaly_scores()` / `.dependency_network()` / `.factors()` / `CAFE().forecast()`; no quantity is illustrative.

By-product	Metric	CAFÉ	Baseline	Baseline detector
Anomaly score	ROC-AUC $\uparrow$	0.963	0.991	Rolling Hampel/MAD z
Dependency net	edge AUC $\uparrow$	0.625	0.389	Pearson corr.
Factors	CCA $\uparrow$	0.904	0.998	Raw-data PCA
Forecast ($h=12$)	MAE $\downarrow$	0.591	0.281	Naive last-value

Table 28: **Wall-clock on a fixed real task** (BeijingAir, first 3000 rows $\times$ 132 sensors, 10% MCAR; MAE on standardised data). Single machine, *one CPU core* with BLAS pinned to 2 threads for every method; *no GPU*. Median of 3 runs. CAFÉ matches the references’ accuracy at comparable CPU cost while being the only strictly *causal* (point-in-time) low-error method. Deep neural imputers (SAITS/BRITS/Transformer/CSDI) are omitted here as they require GPU *training*; see text for that $\sim 10^3 \times$ compute gap.

Method	Backbone	Time (s) ↓	MAE ↓	Causal?	HW
CAFÉ (ours)	factor	1.16	0.134	✓	CPU
SoftImpute	low-rank	1.03	0.513	✗	CPU
TRMF	MF+AR	1.73	0.288	✗	CPU
Linear interp	1D interp	0.01	0.155	✗	CPU
kNN	kNN	1.20	0.192	✗	CPU

Table 29: **CAFE at scale** (Gap: prior panels are all mid-size, ≤ 100 channels). CAFÉ run *causally* (`online_impute`) on genuinely wide panels – up to 862 channels, a decade more width than the race – at 10% MCAR. MAE on the standardise-on-observed scale (held-out cells only); single CPU core, BLAS pinned to 2 threads, *no GPU*; median of 2 runs. The measured log–log time-vs- N slope (1.15) confirms CAFÉ stays strongly sub-quadratic in width at this scale. The PhysioNet-2012 ICU row reports CAFE vs simple causal baselines on real clinical native missingness (extra synthetic hold-out for ground truth).

Wide panel	T	N	MAE ↓	Time (s) ↓
Traffic (PEMS), full	2500	862	0.140	15.76
Electricity (UCI), full	2500	321	0.233	2.61
<i>Measured N-scaling slope on Traffic (PEMS), full (log–log, $N=50 \rightarrow 800$):</i> 1.15 (1.0 = linear)				
<i>PhysioNet-2012 ICU (1000 ICU stays $\times$ 48h $\times$ 36 vars, 81% native missing; causal MAE on synthetic hold-out):</i>				
CAFÉ 0.532 / LOCF 0.441 / Running-mean 0.515				

B What CAFÉ yields from one causal pass

Every panel below is produced by the *same* forward run of the model on real or synthetic data and read straight from its internal state—no quantity is illustrative. Together they show that a single causal estimator delivers a portfolio of capabilities that normally require separate tools.

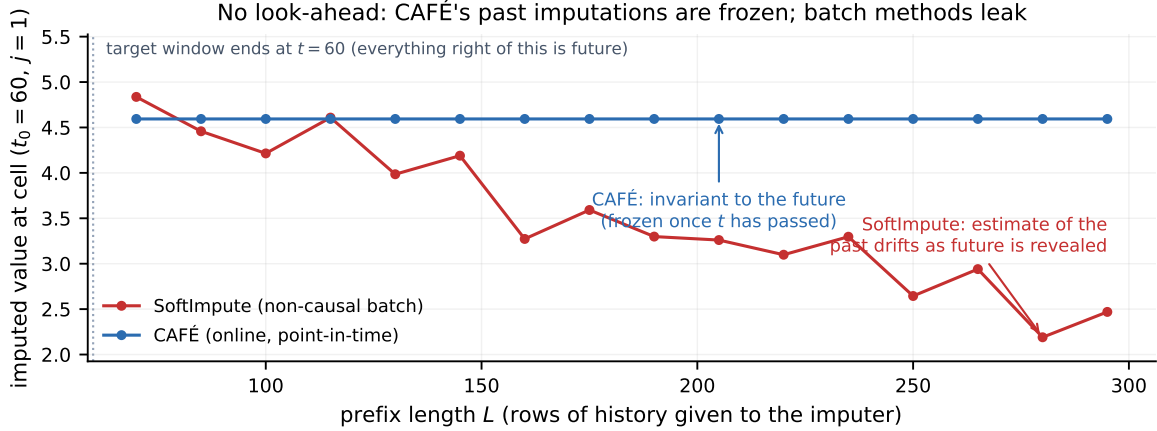


Figure 7: **The moat: no look-ahead.** For a fixed early missing cell, we re-impute on growing time-prefixes. CAFÉ’s estimate is *frozen* the moment its time passes (flat line), so a backtest cannot be contaminated; the non-causal batch method’s “past” estimate keeps changing as the future is revealed. This is what makes CAFÉ safe for sequential decision-making.

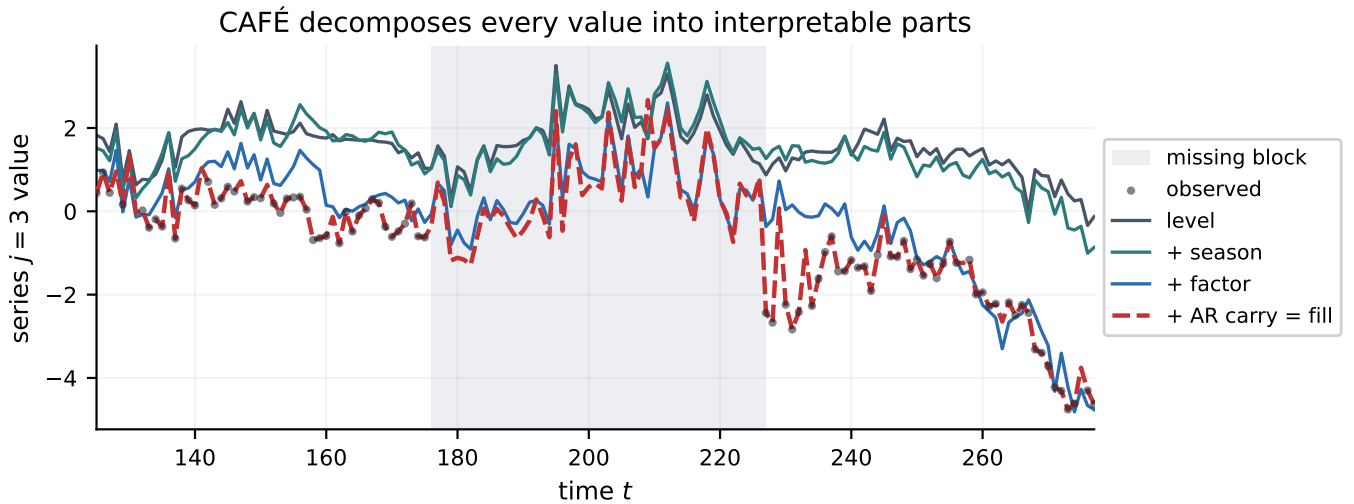


Figure 8: **Interpretability.** The imputed value is the sum of human-readable parts—level, season, shared factors, idiosyncratic carry—read from the trace, not a black box. Each term can be inspected, audited, or used on its own.

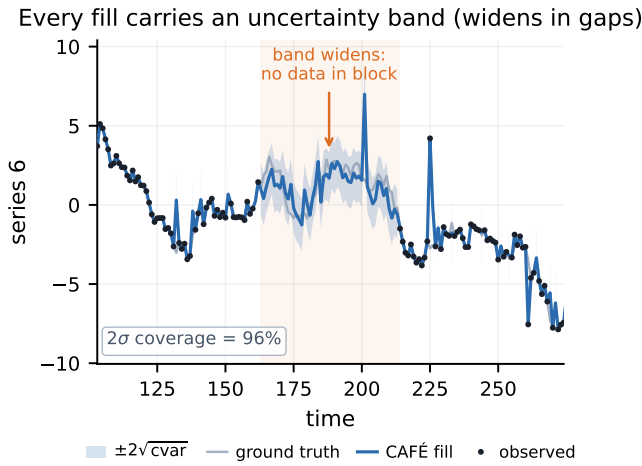


Figure 9: **Conservative uncertainty.** Every fill carries a posterior band that widens inside long gaps (less information). The bands are *predominantly over-conservative*—they over-cover at most levels (a safe failure mode for decision-making), with only marginal undershoots at the tightest levels—and enable confidence intervals, error propagation, and multiple imputation. Scored by CRPS / coverage / sharpness (Table 25).

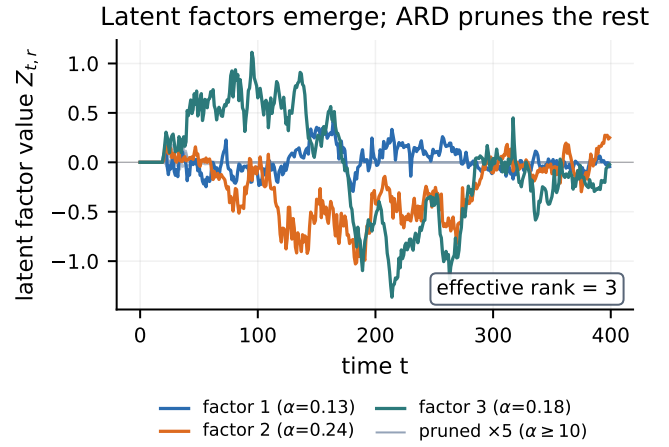


Figure 10: **Latent factors & learned rank.** The common factors z_t (a streaming robust DFM); ARD prunes the rest, so the effective rank is discovered, not set.

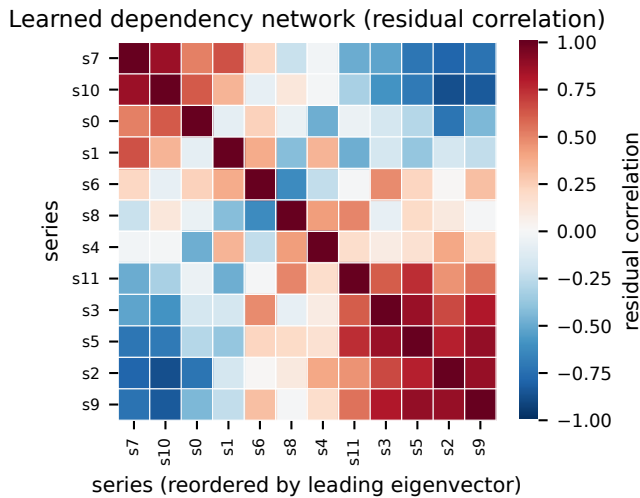


Figure 11: **Dependency network.** The residual covariance recovers the cross-sectional correlation structure between series—useful as features and for contagion analysis.

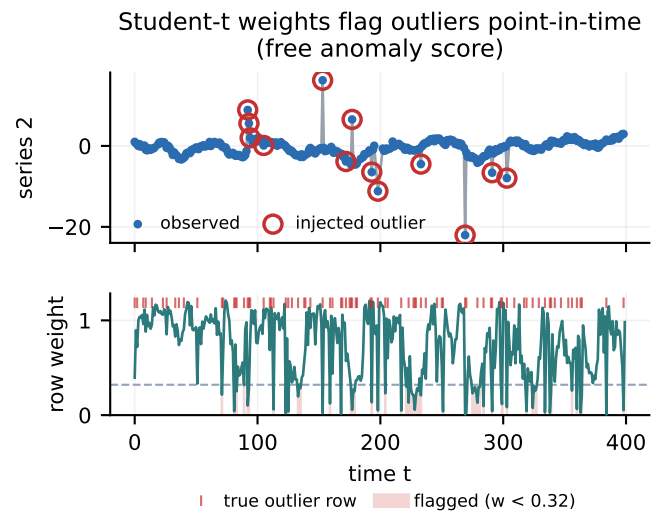


Figure 12: **Anomaly detection (free).** The Student- t weights drop exactly on injected outliers—a point-in-time data-quality / fault score from the same pass.

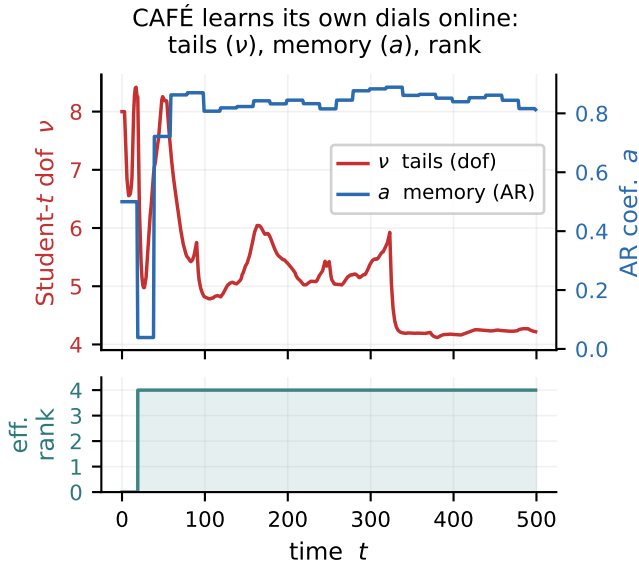


Figure 13: **Self-tuning dials.** The tails (ν), memory (a) and effective rank adapt online from the data—no hyperparameters.

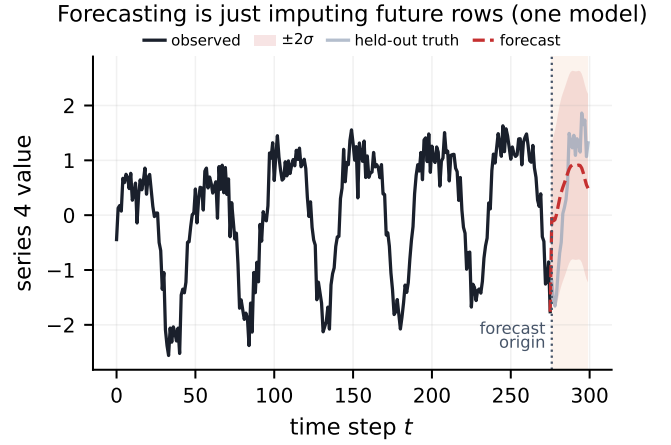


Figure 14: **Forecasting = imputation.** Masking the last rows and imputing them yields a forecast via the AR/Kalman state—one model for fill, forecast and now-cast. The shaded $\pm 2\sigma$ cone is the model's own predictive variance, widening with horizon as the latent state's information decays.

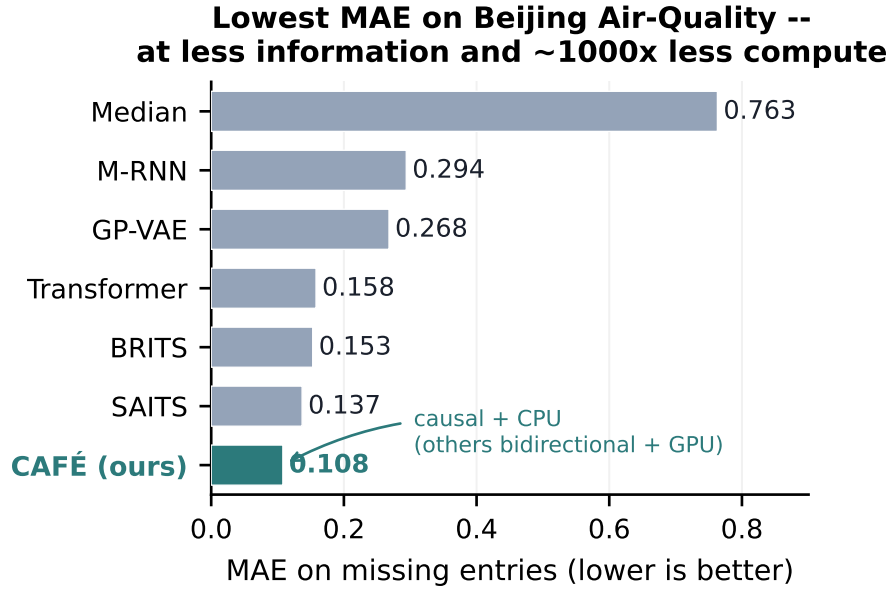


Figure 15: **Accuracy.** CAFÉ's MAE on the named Beijing Air-Quality slice (10% point/MCAR, standardised, 3 seeds) lands in the band of GPU-trained bidirectional deep models—at far less information and compute, and while remaining causal. Published deep numbers (different protocols) are context, not a head-to-head ranking; see Table 11.

C Experimental figures

These panels accompany the experiments of §3; every number is from a real run on the stated data.

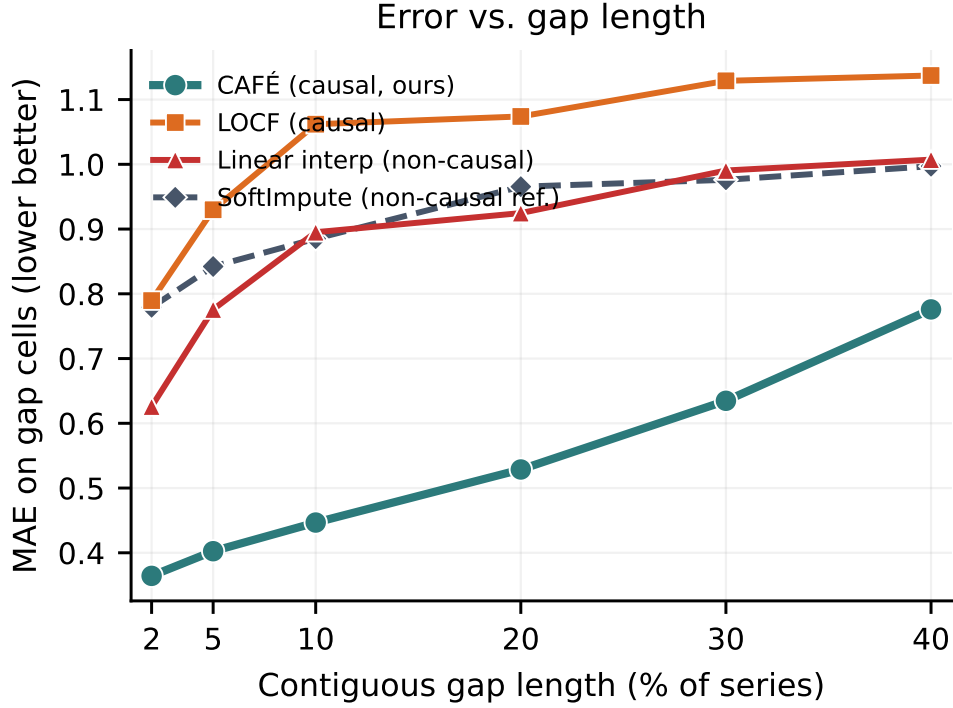


Figure 16: **Graceful degradation under long gaps (Table 21)**. Imputation MAE as a single contiguous gap grows to 40% of the series. Local causal methods (LOCF, linear) saturate near the unconditional error; CAFÉ stays bounded and beats even the non-causal SoftImpute reference at every length.

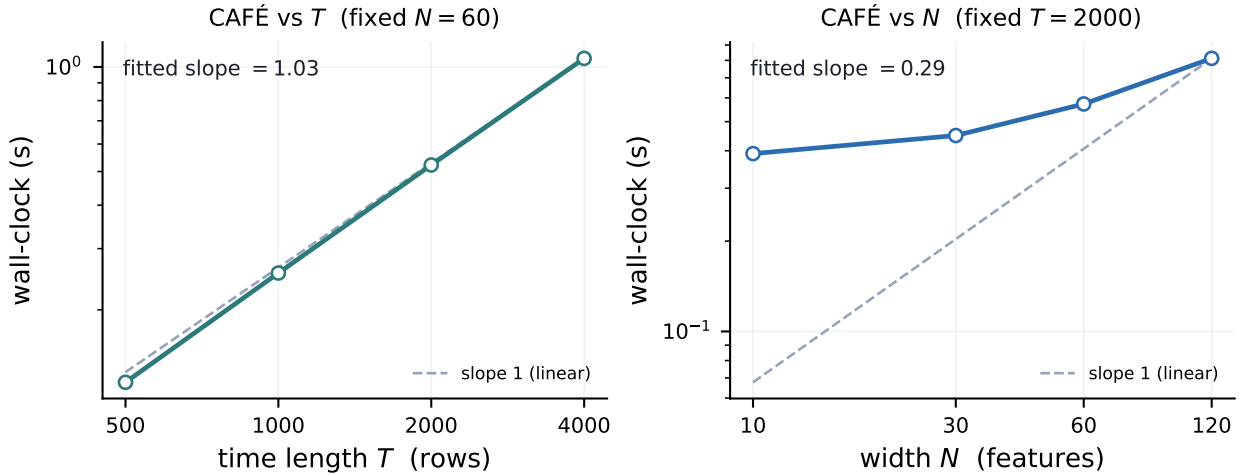


Figure 17: **Scaling (Table 28)**. Wall-clock vs. series length T (near-linear, log-log slope ≈ 1) and vs. width N (strongly sub-linear): each row uses only rank- R Cholesky solves, never an $N \times N$ inverse.

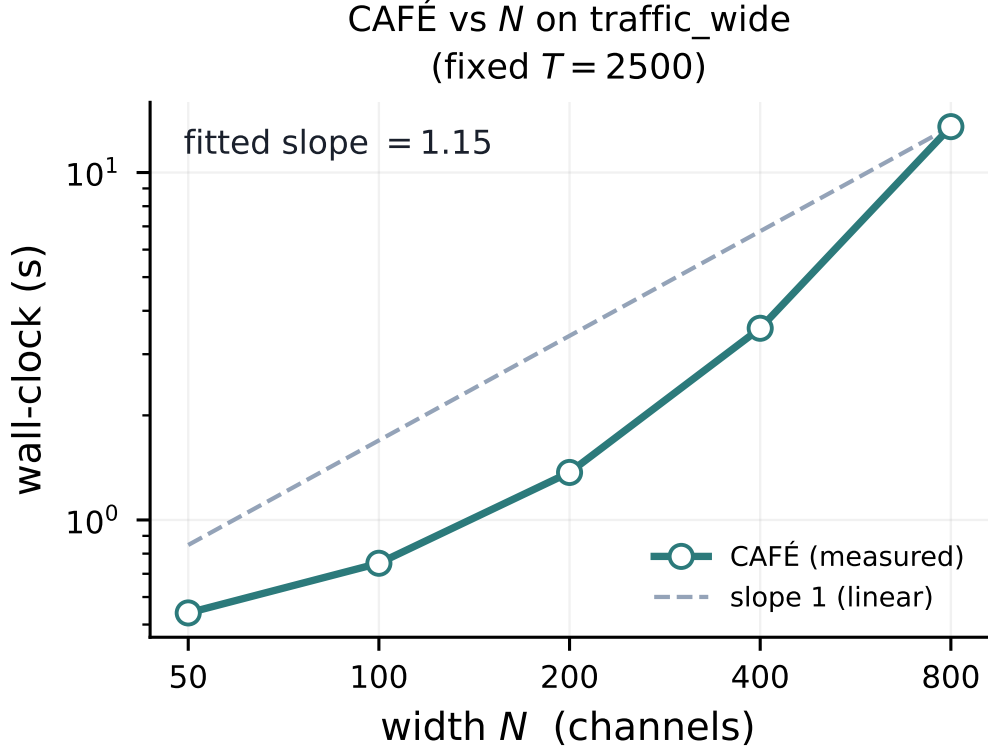


Figure 18: **Scaling at real width** (Table 29). Wall-clock vs. width N on the full 862-sensor Traffic panel (a decade wider than the race panels): the measured log-log slope is ≈ 1.15 , confirming CAFÉ stays near-linear in N at scale.

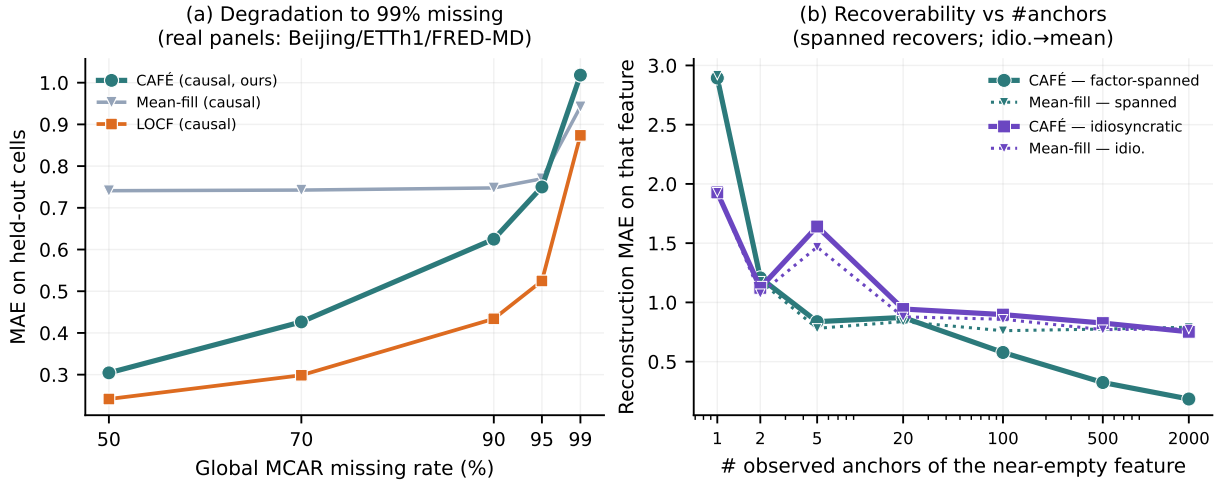


Figure 19: **Extreme missingness and the recoverability frontier** (Table 22). *Left*: MAE under global MCAR up to 99% (causal methods only)—CAFÉ stays finite and bounded below mean-fill, while carry-forward (LOCF) wins under scattered point missingness. *Right*: reconstruction MAE of a single near-empty feature vs. the number of observed anchors k : when the feature is factor-spanned CAFÉ recovers it (margin over mean-fill grows with k), when it is idiosyncratic CAFÉ tracks the mean rather than hallucinating; $k \leq 2$ is the honest identifiability floor.

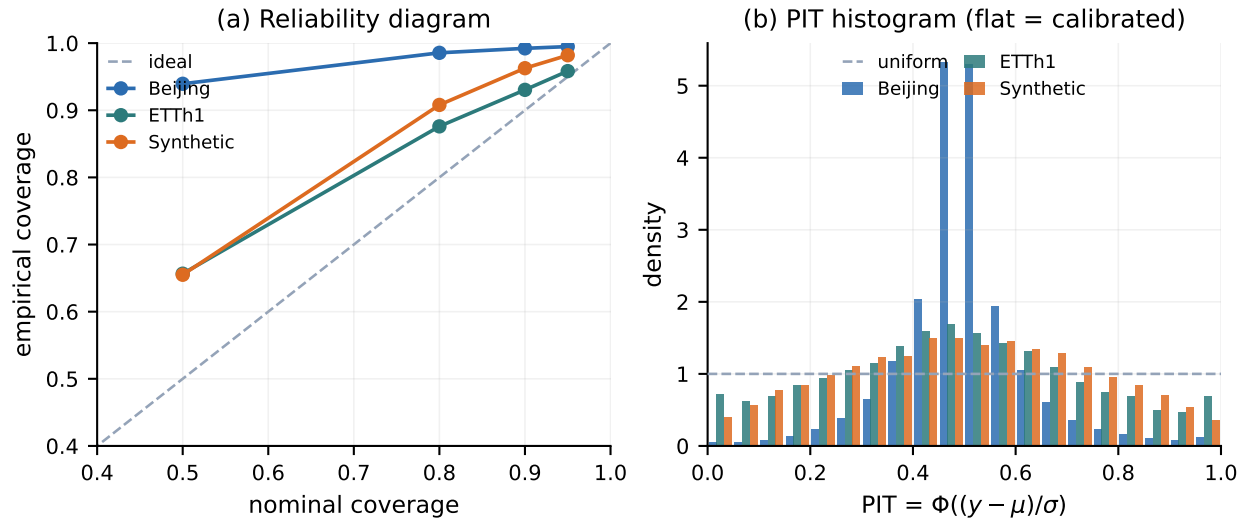


Figure 20: **Calibration** (Table 25). (a) Reliability diagram: empirical vs. nominal coverage—curves sit *on or above* the diagonal at most levels (predominantly over-covering), with only marginal undershoots at the tightest levels. (b) PIT histograms: a flat profile would be sharply calibrated; CAFÉ’s are centre-peaked, the signature of conservative (over-dispersed) intervals—safe, but loose, on the smoother panels.

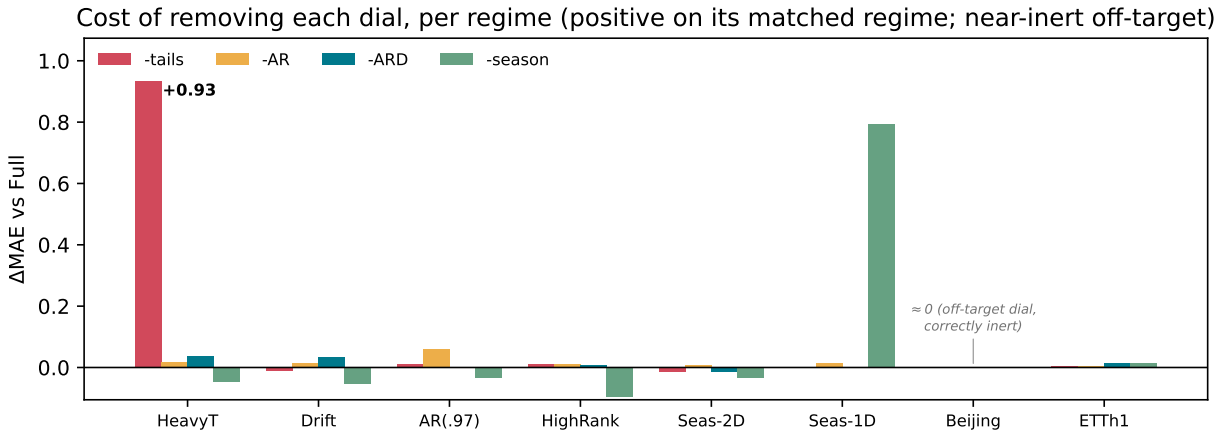


Figure 21: **Full ablation matrix** (Table 19). Cost of removing each dial (ΔMAE vs. full CAFÉ) across all eight regimes. Each dial spikes on the regime it targets and is near-inert—small, often slightly negative—elsewhere, the price of empirical-Bayes priors that shrink an unsupported dial toward off rather than exactly to it.

References

- [1] R. Mazumder, T. Hastie, R. Tibshirani. Spectral regularization algorithms for learning large incomplete matrices. *JMLR*, 2010.
- [2] H.-F. Yu, N. Rao, I. Dhillon. Temporal regularized matrix factorization for high-dimensional time series prediction. *NeurIPS*, 2016.
- [3] A. C. Harvey. *Forecasting, Structural Time Series Models and the Kalman Filter*. Cambridge Univ. Press, 1989.
- [4] S. Athey, M. Bayati, N. Doudchenko, G. Imbens, K. Khosravi. Matrix completion methods for causal panel data models. *JASA*, 2021.
- [5] W. Cao et al. BRITS: Bidirectional recurrent imputation for time series. *NeurIPS*, 2018.
- [6] W. Du, D. Côté, Y. Liu. SAITS: Self-attention-based imputation for time series. *Expert Systems with Applications*, 2023.
- [7] Y. Tashiro, J. Song, Y. Song, S. Ermon. CSDI: Conditional score-based diffusion models for probabilistic time series imputation. *NeurIPS*, 2021.
- [8] A. Cini, I. Marisca, C. Alippi. Filling the gaps: multivariate time series imputation by graph neural networks. *ICLR*, 2022.
- [9] C. M. Bishop. Bayesian PCA. *NeurIPS*, 1999.
- [10] A. Vaswani et al. Attention is all you need. *NeurIPS*, 2017.
- [11] V. Fortuin et al. GP-VAE: deep probabilistic time series imputation. *AISTATS*, 2020.
- [12] J. Yoon, W. R. Zame, M. van der Schaar. Estimating missing data in temporal data streams using multi-directional recurrent neural networks (M-RNN). *IEEE TBME*, 2019.
- [13] M. Khayati, A. Lerner, Z. Tymchenko, P. Cudré-Mauroux. Mind the gap: an experimental evaluation of imputation of missing values techniques in time series. *PVLDB* 13(5), 2020.
- [14] W. Du et al. TSI-Bench: benchmarking time series imputation. *arXiv:2406.12747*, 2024. (PyPOTS)
- [15] T. Nie, G. Qin, W. Ma, Y. Mei, J. Sun. ImputeFormer: low rankness-induced transformers for generalizable spatiotemporal imputation. *KDD*, 2024. (arXiv:2312.01728)
- [16] Authors of FGTL. Frequency-domain generative time-series imputation via diffusion. *NeurIPS*, 2024.
- [17] M. Liu, H. Huang, H. Feng, L. Sun, B. Du, Y. Fu. PriSTI: a conditional diffusion framework for spatiotemporal imputation. *ICDE*, 2023.
- [18] S. Fang, Q. Wen, Y. Luo, S. Zhe, L. Sun. BayOTIDE: Bayesian online multivariate time series imputation with functional decomposition. *ICML*, 2024.
- [19] Y. Zhao, E. Landgrebe, E. Shekhtman, M. Udell. Online missing value imputation and change point detection with the Gaussian copula. *AAAI*, 2022.
- [20] L. Balzano, R. Nowak, B. Recht. Online identification and tracking of subspaces from highly incomplete information (GROUSE). *Allerton*, 2010.
- [21] C. Ma, C. Zhang. Identifiable generative models for missing not at random data imputation. *NeurIPS*, 2021.
- [22] Cheng et al. A causal view of time series imputation. *IJ-CAI*, 2025.
- [23] S. Bryzgalova, S. Lerner, M. Lettau, M. Pelger. Missing financial data. *Review of Financial Studies*, 2025.
- [24] X. Chen, C. Zhang, X.-L. Zhao, N. Saunier, L. Sun. Non-stationary temporal matrix factorization for multivariate time series forecasting (NoTMF). *arXiv:2203.10651*, 2022.
- [25] K. Gilman, D. Hong, J. A. Fessler, L. Balzano. Streaming heteroscedastic probabilistic PCA with missing data (SHASTA-PCA). *arXiv:2310.06277*, 2023.
- [26] J. He, L. Balzano, A. Szlam. Incremental gradient on the Grassmannian for online foreground/background separation in subsampled video (GRASTA; robust GROUSE). *CVPR*, 2012. *arXiv:1109.3827*.
- [27] K. Obata, K. Kawabata, Y. Matsubara, Y. Sakurai. Mining of switching sparse networks for missing value imputation in multivariate time series (MissNet). *KDD*, 2024. *arXiv:2409.09930*.
- [28] M. Zaffran, A. Dieuleveut, J. Josse, Y. Romano. Conformal prediction with missing values. *ICML*, 2023. *arXiv:2306.02732*.
- [29] Y. Gui, R. Foygel Barber, C. Ma. Conformalized matrix completion. *NeurIPS*, 2023. *arXiv:2305.10637*.
- [30] M. Goswami, K. Szafer, A. Choudhry, Y. Cai, S. Li, A. Dubrawski. MOMENT: a family of open time-series foundation models. *ICML*, 2024. *arXiv:2402.03885*.
- [31] E. Fons et al. MoTM: towards a foundation model for time series imputation based on continuous modeling. *AALTD@ECML-PKDD*, 2025. *arXiv:2507.13207*.
- [32] S. B. Hoo, S. Müller, D. Salinas, F. Hutter. From tables to time: extending TabPFN-v2 to time series forecasting (TabPFN-TS). *arXiv:2501.02945*, 2025.
- [33] Are time-indexed foundation models the future of time series imputation? *arXiv:2511.05980*, 2025.
- [34] J. Bai, S. Ng. Determining the number of factors in approximate factor models. *Econometrica*, 70(1):191–221, 2002.
- [35] M. Barigozzi, L. Trapin. Estimation of large approximate dynamic matrix factor models based on the EM algorithm and Kalman filtering. *arXiv:2502.04112*, 2025.
- [36] S. Liu, X. Li, G. Cong, Y. Chen, Y. Jiang. Multivariate time-series imputation with disentangled temporal representations (TIDER). *ICLR*, 2023.
- [37] X. Chen, M. Lei, N. Saunier, L. Sun. Low-rank autoregressive tensor completion for spatiotemporal traffic data imputation (LATC). *IEEE T-ITS*, 2022. *arXiv:2104.14936*.
- [38] R. E. Kalman. A new approach to linear filtering and prediction problems. *Trans. ASME J. Basic Eng.*, 82(1):35–45, 1960.
- [39] H. E. Rauch, F. Tung, C. T. Striebel. Maximum likelihood estimates of linear dynamic systems. *AIAA Journal*, 3(8):1445–1450, 1965.

- [40] B. D. O. Anderson, J. B. Moore. *Optimal Filtering*. Prentice-Hall, 1979.
- [41] R. El-Yaniv, Y. Wiener. On the foundations of noise-free selective classification. *JMLR*, 11:1605–1641, 2010.
- [42] Y. Geifman, R. El-Yaniv. Selective prediction with deep neural networks. *NeurIPS*, 2017. *arXiv:1705.08500*.
- [43] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, T. Schuster. Conformal risk control. *ICLR*, 2024. *arXiv:2208.02814*.
- [44] D. Giannone, L. Reichlin, D. Small. Nowcasting: the real-time informational content of macroeconomic data. *Journal of Monetary Economics*, 55(4):665–676, 2008.
- [45] R. S. Mariano, Y. Murasawa. A new coincident index of business cycles based on monthly and quarterly series. *Journal of Applied Econometrics*, 18(4):427–443, 2003.
- [46] E. Ghysels, P. Santa-Clara, R. Valkanov. The MIDAS touch: mixed data sampling regression models. Working paper, 2004.
- [47] M. Bańbura, D. Giannone, M. Modugno, L. Reichlin. Now-casting and the real-time data flow. In *Handbook of Economic Forecasting*, vol. 2A, pp. 195–237, Elsevier, 2013.